

A FINITELY PRESENTED NON-MF GROUP

ANONYMOUS

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ABSTRACT. A countable discrete group is MF in the Carrión–Dadarlat–Eckhardt–Shulman sense if it embeds into the unitary group of a norm matrix corona $\prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C})$. As of August 2026, whether every group is MF remained open. The precise group notion dates to 2012–2013 and has an operator-algebraic lineage going back to the Blackadar–Kirchberg MF program of 1997. It is one of a family of metric approximation problems whose normalized-Hamming (sofic) case was solved recently by OpenAI and subsequently generalized by Kun and Thom, and whose unnormalized Frobenius case was solved by De Chiffre, Glebsky, Lubotzky and Thom. We construct an explicit finitely presented group E , together with a designated central involution $w \in E$, such that $w \neq 1$ but *every* homomorphism from E to the unitary group of every norm matrix corona of the displayed form kills w . Consequently E is not MF, and neither its full nor its reduced group C^* -algebra is an MF algebra. The group is built from $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$, one HNN letter implementing the proper endomorphism $(v, A) \mapsto (2v, A)$, and one additional involution; the proof combines the Kazhdan projection of Γ with stable finiteness of matrix coronas. The finite-dimensional case of the kill theorem is elementary and uses neither property (T) nor unitarity; we also show, via quasidiagonality of amenable groups, that the same marked presentation with a cyclic base can have a corona representation in which the marked involution survives. The kill theorem is proved as a general finite-normal criterion: if a countable group contains the image of a property (T) group, compressed by a letter t and centralized by a letter c , then every finite normal subgroup contained in the compression-defect subgroup — the normal closure of the commutators of tct^{-1} with that image — is killed by every homomorphism into the unitary group of every C^* -subalgebra of a norm matrix corona. More generally, finiteness can be replaced by property (T): every normal property (T) subgroup contained in the compression defect lies in the MF radical, and may be infinite, noncentral, and torsion-free. Among the consequences: $C_{\mathrm{red}}^*(E)$ is a unital separable C^* -algebra with a faithful tracial state — hence stably finite — that is not an MF algebra, giving an explicit reduced-group- C^* -algebra realization of the already known failure of stable finiteness to imply the MF property; MF groups are closed under neither quotients nor split extensions; every countable group has a largest MF quotient, by its MF radical, and the radical element w is portable: any group receiving E with w surviving is non-MF; for every $m \geq 2$ a scaled variant E_m ($E = E_2$) is a finitely presented non-MF group, and there exist $2^{\aleph_0}$ pairwise non-isomorphic finitely generated non-MF groups; the MF locus is closed in every fixed-rank marked-group space, equivalently non-MF-ness is open; and the kill theorem holds in a uniform quantitative form over all matrix sizes at once.

1. INTRODUCTION

Let $(d_n)_{n \in \mathbb{N}}$ be a sequence of positive integers and let

$$(1) \quad \mathcal{Q} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C})$$

be the associated *norm matrix corona*: bounded sequences of matrices modulo sequences converging to 0 in operator norm. Following Blackadar–Kirchberg [BK], a separable C^* -algebra is *MF* if it embeds into some algebra of the form (1). We use *MF group* exclusively in the sense introduced by Carrión–Dadarlat–Eckhardt and used by Shulman [CDE, Sh]: a countable discrete group E is MF if there is an *injective* group homomorphism

$$E \longrightarrow \mathcal{U}(\mathcal{Q})$$

for some choice of (d_n) . Equivalently, E admits finite-dimensional approximate unitary representations that are approximately injective, with all defects measured in operator norm. Carrión–Dadarlat–Eckhardt take the dimensions to be increasing. Allowing an arbitrary positive sequence, as we do, is equivalent: at stage n , take the block sum of the first n models and place each old unitary in its corresponding block. Multiplicative defects are the maxima of the block defects, while every separation already obtained persists at all later stages. This qualification matters: Shulman notes that a strictly stronger group property has also been called MF [Sh, Definition 3 and footnote 1]: *purely* MF requires the approximating maps to be genuine finite-dimensional representations. The two properties must not be conflated when comparing with the literature: Magee–de la Salle proved that $\mathrm{SL}_4(\mathbb{Z})$ is not purely MF [MdS], while $\mathrm{SL}_4(\mathbb{Z})$ is residually finite and hence MF in the sense used here. Residually finite groups are MF, amenable groups are MF as a consequence of the Tikuisis–White–Winter theorem [TWW], free products of amenable groups are MF, and a substantial permanence theory exists [Sh].

The age of the problem has two layers. Blackadar and Kirchberg introduced MF C^* -algebras in 1997 [BK]; the broader stably-finite-to-MF embedding question attributed to Kirchberg implies a positive answer to the corresponding operator-norm approximation problem for groups [DGLT, Introduction]. The precise group property used here was introduced as Definition 2.7 by Carrión–Dadarlat–Eckhardt in a preprint first posted on October 15, 2012 and published in 2013 [CDE]. Their paper introduces the definition but does not formulate the universal group question in those words, so we distinguish the 1997 problem lineage from the 2012–2013 definition. At the date of the present result, these were respectively about twenty-nine years and nearly fourteen years old. As late as May 2026, Bachner–Dogon–Lubotzky described whether every group is MF as a longstanding open problem [BDL, pp. 235–238]. It is the operator-norm member of the family of approximation problems surveyed in [DGLT, BDL], whose unnormalized-Frobenius case was solved negatively in [DGLT]. Building on rigidity theorems for sofic approximations of Kazhdan groups due to Kun [Ku16] and Kun–Thom [KT19], OpenAI constructed the first nonsofic group [OAI]; Kun and Thom then extracted and generalized the mechanism, obtaining nonsofic generalized wreath products [KT], and Fournier-Facio exhibited finitely presented torsion-free examples [FF]. In the sofic problem the metric is normalized Hamming

distance on permutation groups, not rank distance; the three metric conventions are stated together in [Sh, Introduction]. Neither result supplies an operator-norm conclusion: [DGLT] treats the unnormalized Frobenius norm, while permutation/Hamming-metric arguments do not directly control operator-norm defects.

Against this August 2026 literature record, the theorem below answers the question negatively with an explicit finitely presented example.

Theorem A (non-MF group). *Let $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$, let $\alpha: \Gamma \rightarrow \Gamma$ be the injective endomorphism $\alpha(v, A) = (2v, A)$, and let $a = (e_1, 1) \in \Gamma$, which does not lie in $\alpha(\Gamma)$. Let E be the finitely presented group obtained from Γ by adjoining a stable letter t with $t\gamma t^{-1} = \alpha(\gamma)$ for $\gamma \in \Gamma$, an involution c commuting with Γ , and by imposing that*

$$w := [tct^{-1}, a(tct^{-1})a^{-1}]$$

is central and satisfies $w^2 = 1$ (Definition 3.2). Then:

- (1) *$w \neq 1$ in E , so w is a central involution;*
- (2) *for every sequence (d_n) and every group homomorphism $\Theta: E \rightarrow \mathcal{U}(\mathcal{Q})$ into the corresponding norm matrix corona (1), one has $\Theta(w) = 1$.*

In particular E is not MF, and neither $C_{\max}^(E)$ nor $C_{\mathrm{red}}^*(E)$ is an MF C^* -algebra.*

A completely explicit presentation of E — eight generators, with every relation displayed — appears as (6) in Definition 3.2. In particular, finitely presented non-MF groups exist.

The proof turns a C^* -algebraic finiteness argument into a group-level kernel theorem. For orientation, it can be organized into the following five steps. The individual ingredients and the surrounding compression/lamp architecture have substantial antecedents. What is new here is the specific stable-finite Kazhdan pinning argument in a norm matrix corona and its conversion into a universal marked kernel.

- (1) *Central corner.* If $\Theta(w) \neq 1$, then $e = \frac{1}{2}(1 - \Theta(w))$ is a nonzero projection commuting with the image of Θ ; cutting by a lift of e replaces $\mathcal{Q}$ by another norm matrix corona on which $\Theta(w) = -1$ (Lemma 6.5). This cut eliminates small-relative-rank phenomena: on the corner, the deviation of w from 1 has full normalized size. This is a norm-corona adaptation of the negative-eigenspace compression for a central involution in the proof of [SV, Proposition 2.7]; the distinction is that no positive relative-rank estimate is needed here.
- (2) *Conjugation representation.* Unitary lifts $U_{g,n}$ of the corner representation conjugate the corner coordinates $M_{r_n}(\mathbb{C})$, and exact multiplicativity of Θ in the quotient makes arbitrary coordinate lifts asymptotically multiplicative in operator norm. Thus $g \mapsto (\mathrm{Ad} U_{g,n})_n$ defines an *exact* homomorphism β into the norm corona of the algebras $B(L^2(M_{r_n}))$ (Lemma 7.9).

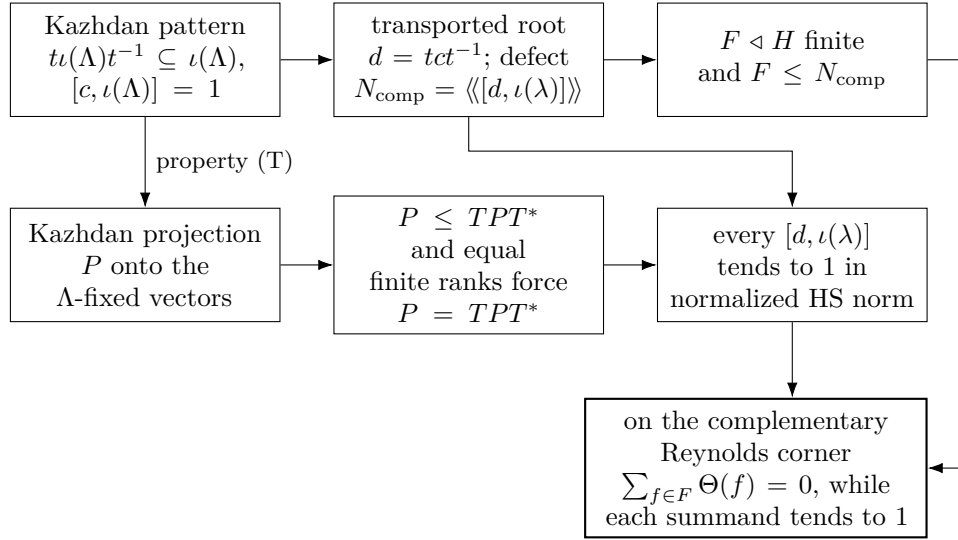


FIGURE 1. The finite-normal obstruction. The upper lane is group-theoretic and the lower lane is the finite-matrix rigidity input. Stable finiteness is used only in the middle lower box; property (T) is used only to make the fixed-space projection uniform across the matrix sequence.

- (3) *Kazhdan projection.* Γ has property (T), so its maximal group C^* -algebra contains the Kazhdan projection p_Γ , whose image under every representation is the projection onto the invariant vectors. The HNN relation $t\Gamma t^{-1} = \alpha(\Gamma) \subseteq \Gamma$ forces, in *every* C^* -representation, the order relation $P \leq \beta(t)P\beta(t)^*$ between the images of p_Γ and its t -conjugate (Lemma 7.10).
- (4) *Stable finiteness.* A norm matrix corona is stably finite: isometries in every matrix amplification are unitaries (Lemma 6.3). Hence the two unitarily equivalent projections in step (3) must be equal: $\beta(t)$ *preserves* the Γ -fixed space.
- (5) *The marked word.* The involution c commutes with Γ , so the matrix vector representing $\Theta(c)$ is Γ -fixed; by step (4) so is the vector representing $\Theta(tct^{-1})$; hence $a \in \Gamma$ asymptotically fixes it, and the commutator word w is forced to 1 in normalized Hilbert–Schmidt norm along the chosen ultrafilter — while the corner normalization of step (1) says it equals -1 in operator norm. The two conclusions contradict each other (Section 8).

Figure 1 shows where the two halves of the proof meet. The upper lane is purely group-theoretic; the lower lane is the finite-matrix rigidity input. Their common output is incompatible with the Reynolds corner.

Nontriviality of w is witnessed by a completely explicit representation of E into a semidirect product of a Clifford group by a linear group over $\mathbb{Z}[1/2]$ (Section 4); no combinatorial theory of HNN extensions is needed.

Two companion results clarify the mechanism. First, in finite dimensions the kill theorem is elementary and needs neither property (T), nor unitarity, nor centrality of w :

For a subgroup $L \leq H$, put

$$\mathcal{C}(L) = \{s \in H : sLs^{-1} \subseteq L\}, \quad G_{\text{comp}}(L) = \langle \mathcal{C}(L) \rangle,$$

and define the intrinsic *compression-centralizer defect*

$$(2) \quad \mathfrak{D}(H, L) = \left\langle \left\langle [gzg^{-1}, \ell] : g \in G_{\text{comp}}(L), z \in C_H(L), \ell \in L \right\rangle \right\rangle_H.$$

Theorem B (finite-dimensional obstruction). *Let k be a field, V a finite-dimensional k -vector space, H a group, $\Gamma \leq H$ a subgroup, and $t, c \in H$, $a \in \Gamma$ elements with $t\Gamma t^{-1} \subseteq \Gamma$ and $c\gamma = \gamma c$ for all $\gamma \in \Gamma$. Then every homomorphism $\pi : H \rightarrow \text{GL}(V)$ satisfies*

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = 1.$$

More generally, every such π kills the entire subgroup $\mathfrak{D}(H, \Gamma)$.

Applied to $H = E$, Theorem B shows that *every* finite-dimensional representation of E over *every* field kills w : finite-dimensional linear representations do not separate the points of E , no finite quotient can witness $w \neq 1$, and the infinite Clifford witness of Section 4 cannot be replaced by a finite-dimensional one. Property (T) enters the corona argument for one reason: it makes “the projection onto the Γ -fixed space” an element of the corona, uniformly across all coordinates at once.

Second, the fixed-space rigidity in that argument cannot simply be omitted:

Theorem C (cyclic-base comparison). *Let $G_0 = \text{BS}(1, 2) = \langle \gamma_0, t \mid t\gamma_0 t^{-1} = \gamma_0^2 \rangle$, let $\Gamma_0 = \langle \gamma_0 \rangle \cong \mathbb{Z}$, let $X_0 = G_0/\Gamma_0$, and let $E_0 = \text{ClLamp}(X_0) \rtimes G_0$ be the corresponding Clifford extension (Construction 4.1), with central sign z_0 . Also define the literal marked-presentation analogue*

$$E_{\text{BS}} = \left\langle \gamma_0, t, c \left| \begin{array}{l} t\gamma_0 t^{-1} = \gamma_0^2, \quad c^2 = 1, \quad [c, \gamma_0] = 1, \\ w_{\text{BS}}^2 = 1, \quad [w_{\text{BS}}, \gamma_0] = [w_{\text{BS}}, t] = [w_{\text{BS}}, c] = 1 \end{array} \right. \right\rangle,$$

where $w_{\text{BS}} = [tct^{-1}, \gamma_0(tct^{-1})\gamma_0^{-1}]$. Then E_0 is an amenable MF group and $z_0 \neq 1$. There is a quotient homomorphism $E_{\text{BS}} \twoheadrightarrow E_0$ sending w_{BS} to z_0 ; consequently the literal analogous presentation has a corona representation in which its marked involution survives. Nevertheless, every finite-dimensional representation of either group kills the corresponding marked element.

The quotient statement makes the comparison at the level actually used by the proof: the marked-word relations and witness mechanism survive when the Kazhdan base is replaced by $\mathbb{Z}$, but the universal corona-kill conclusion does not. Thus property (T), or a substitute furnishing a uniform fixed-space

projection, is essential to this argument; this is not a claim that every possible proof of the concrete Theorem A must invoke property (T).

Generalizations and consequences. The corona half of Theorem A is proved once, in the following generality (Definition 7.1). A *marked Kazhdan pattern* in a countable group H is a tuple (Λ, ι, t, c) consisting of a countable group Λ with property (T), a homomorphism $\iota: \Lambda \rightarrow H$ — injectivity is *not* assumed — and elements $t, c \in H$ with $t\iota(\Lambda)t^{-1} \subseteq \iota(\Lambda)$ and $[c, \iota(\Lambda)] = 1$. Writing $d = tct^{-1}$, the *compression-defect subgroup* of the pattern is the normal closure

$$N_{\text{comp}} = \langle\langle [d, \iota(\lambda)] : \lambda \in \Lambda \rangle\rangle_H.$$

The *finite-normal obstruction criterion* (Theorem 7.2) states: every finite normal subgroup of H contained in N_{comp} is killed by every corona representation of H . Theorem 7.3 replaces the finite subgroup by an arbitrary normal property (T) subgroup and places that entire subgroup in the MF radical. Theorem A(2) is the instance $H = E$ with $F = \{1, w\}$: since $c^2 = 1$ forces $d^2 = 1$, the marked word satisfies the identity $w = [d, a]^2 \in N_{\text{comp}}$ (Lemma 8.1). The corner cut that starts the proof is the finite-group averaging (Reynolds) projection $|F|^{-1} \sum_{f \in F} \Theta(f)$, whose complement for $F = \{1, w\}$ is exactly the central corner of step (1) above. Corollary 9.5 upgrades the target from coronas to arbitrary C^* -subalgebras of coronas, so that, for instance, every homomorphism from H into the unitary group of a unital separable quasidiagonal C^* -algebra, or of $C_{\text{red}}^*(F_k)$ for a free group F_k [HT], kills such an F . Theorem C calibrates the hypotheses: for the same pattern over the base $\mathbb{Z}$, the conclusion fails.

Section 9 records consequences in four directions.

(i) *C^* -algebras.* Blackadar and Kirchberg proved that MF algebras are stably finite and asked about the converse: whether every stably finite separable C^* -algebra is MF [BK], a question sometimes described — in Kirchberg’s stronger formulation, a conjecture — as a C^* -analogue of the Connes embedding problem (see, e.g., the introduction of [Ra]). The converse was already known to fail after the negative solution of the Connes embedding problem from $\text{MIP}^* = \text{RE}$ [JNVWY]: Goldbring gives the direct deduction by passing from a non- $\mathcal{R}^\omega$ -embeddable II_1 factor to a separable stably finite non-MF C^* -algebra [Go23, Corollary 5.2]. The new point here is that the counterexample is the reduced group C^* -algebra of the explicit finitely presented group constructed above. Indeed, as recently as March 2026 no full or reduced group C^* -algebra was known not to be MF [Sh, Introduction]:

Theorem D (a stably finite non-MF reduced group algebra). *$C_{\text{red}}^*(E)$ is a unital separable C^* -algebra which carries a faithful tracial state — and is therefore stably finite — but is not an MF algebra.*

Here $C_{\text{red}}^*(E)$ is not nuclear (E contains an isomorphic copy of Γ by Corollary 4.5, and Γ is not amenable), so the quasidiagonality problems for *nuclear* stably finite algebras around [BK, TWW] are untouched (Remark 9.8).

In the opposite direction, for the ambient HNN extension G of (5) the algebra $C_{\max}^*(G)$ contains an explicit proper isometry — so it is not finite, in particular not MF — although the group G itself is MF, being residually finite by the Borisov–Sapir mapping-torus theorem [BoS] (Proposition 9.10).

(ii) *Permanence.* MF groups are closed under neither quotients nor split extensions: E is a quotient of the MF group F_8 , and the witness group $W = \text{CILamp}(\bar{X}) \rtimes \bar{G}$ of Section 4 is a finitely generated non-MF group that is a split extension of the residually finite linear group $\bar{G}$ by the locally finite group $\text{CILamp}(\bar{X})$ (Corollaries 9.12 and 9.14); both statements contrast with the positive permanence theory of [Sh].

(iii) *Structure.* The *MF radical* of a countable group — the intersection of the kernels of its corona representations, cf. the finite and finite-approximate quotients of [Slo17] — is functorial under arbitrary homomorphisms, and the quotient by it is the largest MF quotient, realized by a single corona representation (Lemma 9.2 and Proposition 9.3; the structural statement is presumably folklore, and we claim no novelty for it). Portability of the radical element w has a striking consequence: embedding E into a 2-generator simple group S , which is possible by Goryushkin’s theorem [Go], gives a 2-generator simple group all of whose homomorphisms into MF groups are trivial (Corollary 9.15).

(iv) *Multiplicity and uniformity.* For every $m \geq 2$ the construction with the endomorphism $(v, A) \mapsto (mv, A)$ yields a finitely presented non-MF group E_m with $E = E_2$ (Theorem 9.19); the groups $E \times \mathbb{Z}^k$, $k \in \mathbb{N}$, are pairwise non-isomorphic finitely presented non-MF groups, and there exist $2^{\aleph_0}$ pairwise non-isomorphic finitely generated non-MF groups (Corollary 9.22). Finally, the kill theorem is uniform over all matrix sizes at once (Theorem 9.26): for every $\varepsilon > 0$ there are $\delta > 0$ and a finite $F_0 \subseteq E$ such that every map $\varphi: E \rightarrow \mathcal{U}(M_r(\mathbb{C}))$, for any r , with $\|\varphi(g)\varphi(h) - \varphi(gh)\| \leq \delta$ for all $g, h \in F_0$, satisfies $\|\varphi(w) - 1\| < \varepsilon$; the proof is a compactness reduction to Theorem 7.2 and yields no explicit pair (δ, F_0) .

What this paper does not prove. The proof of Theorem A does *not* show that E is nonhyperlinear; equivalently, it does not show that the canonical group trace, or the associated group von Neumann algebra, fails the relevant Connes-embedding property. It also does not touch the question whether hyperlinear groups are sofic. The finiteness step (4) is special to operator-norm coronas: in a tracial (Hilbert–Schmidt) matrix ultraproduct the available conjugation representation acts on L^2 of the ultraproduct. When that Hilbert space is infinite-dimensional, its full operator algebra is properly infinite, so the argument no longer forces nested unitarily equivalent fixed-space projections to be equal. Section 11 makes this precise. The same distinction explains why the group $E/\langle w \rangle$ remains untouched by our argument; the ambient group G is in fact MF (Proposition 9.10). Theorem D likewise concerns only the operator-norm side: it does not bear on quasidiagonality of nuclear

stably finite algebras, nor on tracial (Connes-embedding) approximation questions.

External inputs. The analytic kernel argument in Theorem A uses two non-elementary external inputs beyond elementary linear algebra and C^* -basics: property (T) of $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ (Fact 2.1) and the existence and defining property of the Kazhdan projection (Fact 2.2). The advertised explicit presentation additionally uses the Conder–Robertson–Williams presentation [CRW]; the Clifford nontriviality witness is proved below by an elementary twisted-group-algebra model. Theorem C uses both quasidiagonality of amenable reduced group C^* -algebras [TWW] (Fact 2.3) and the standard quasidiagonal-to-corona embedding theorem (Fact 2.4). Theorem B uses no external input. Section 9 uses a handful of classical external inputs, each confined to the single result citing it: Borisov–Sapir residual finiteness of mapping tori of injective endomorphisms of finitely generated linear groups [BoS, Theorem 1.6] (Proposition 9.10; note that the linear case of [BoS], unlike the free case, relies on an unpublished result of Hrushovski, as its authors state); Mal’cev’s residual finiteness of finitely generated linear groups [Ma] (Corollary 9.14); B. H. Neumann’s continuum family of 2-generator groups [Neu] (Corollary 9.22); and Goryushkin’s embedding of countable groups into 2-generator simple groups [Go] (Corollary 9.15). The Haagerup–Thorbjørnsen theorem that $C_{\mathrm{red}}^*(F_k)$ is MF [HT] is cited only to populate the list of example targets in Corollary 9.5. The proof of Theorem D needs no input beyond Theorem A: the faithfulness of the canonical trace and the resulting stable finiteness are proved inline (Lemma 9.7). Its priority context uses the earlier negative solution of the general MF problem [Go23, Corollary 5.2], ultimately derived from $\mathrm{MIP}^* = \mathrm{RE}$ [JNVWY]. Remark 9.8 cites [BHV, BO] for context only.

Relation to prior work. The closest conceptual analogue is the recent work on nonsofic groups. The group-level skeleton in OpenAI’s Proposition 2.3 consists of a property-(T) subgroup Γ , one-sided containments $t_i \Gamma t_i^{-1} \leq \Gamma$, and a subgroup centralizing Γ that is moved by a compressor [OAI, Proposition 2.3]. Contemporaneous work of Kun–Thom isolates this mechanism through the compression semigroup $P_\Gamma = \{g : g\Gamma g^{-1} \leq \Gamma\}$, infranormality, normalization of centralizers, invariance of fixed algebras, and a sofic Mautner envelope [KT, Theorems A, B and C]. The property-(T) rigidity engine behind these developments — almost-invariant structure for expander sofic approximations of Kazhdan groups, upgraded and propagated — originates in Kun’s expander decomposition [Ku16] and in Kun–Thom [KT19]. Contemporaneously with [OAI, KT], and completed beforehand (see the dated footnote in [AT]), Alekseev–Thom proved that the centralizer of any sofic embedding of a Kazhdan group is a metric ultraproduct of permutation groups, with LEF and residual-finiteness consequences [AT, Theorem A]. All of these results live on the Hamming/tracial side: they extract structural information about the approximations themselves. Lemmas 7.9, 7.10 and 7.11 below instead

run the conjugation representation of operator-norm lifts and extract an exact equality of Kazhdan projections from stable finiteness; the output is not structure of an approximation but a universal marked kernel element. It also records the finite analogue: a one-sided inclusion $t\Gamma t^{-1} \leq \Gamma$ becomes an equality in every finite quotient. We therefore make no novelty claim for the one-sided-compression/lamp or Mautner-type fixed-space-propagation paradigms. In the reverse direction the divergence is equally sharp: the lamp groups of [KT] are plain abelian sums $\bigoplus_{G/\Gamma} \mathbb{Z}/2$, with no central extension, no distinguished sign, and no marked element, and the OpenAI construction uses a commuting Thompson-type witness subgroup with no Clifford structure. The marked-central-kernel device and the Clifford sign witness of the present paper are not adaptations of either.

The base group itself also has a lineage. A finitely presented Kazhdan group carrying a proper “scaling” endomorphism of an abelian normal subgroup belongs to the Cornuier–Abels lineage [Co, Abe]: Thom followed this pattern, with the shift on a central $\mathbb{F}_p[t, t^{-1}]$ - (or $\mathbb{Z}[1/p]$ -)line inside a Kazhdan matrix group, to produce hyperlinear Kazhdan groups that are not residually finite [Th, Section 2]. Our $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ with $\alpha(v, A) = (2v, A)$ is the simplest integral analogue of this scaling pattern (Remark 3.5). The approximation conclusions in [Th] are sofic or hyperlinear and use Hamming or tracial Hilbert–Schmidt models, not operator-norm coronas. Here the replacement for finite cardinality is the implication

$$P \leq \beta(t)P\beta(t)^* \implies P = \beta(t)P\beta(t)^*,$$

because the conjugation representation lands in a stably finite norm matrix corona. The central corner then converts this equality into a fixed nontrivial element killed by every corona representation.

The Clifford and marked-sign sides also have close antecedents. Slofstra uses finite and infinite-rank Clifford groups with a distinguished central involution, a shift semidirect product, and an HNN extension implementing a doubling map in an approximate-representation problem; he also explicitly records an operator-norm variant [Slo18, Section 2 and Remark 3.3]. The placement and purpose of those pieces differ here: the proper compression is in the Kazhdan base, the Clifford group supplies an external nontriviality witness, and the conclusion is a universal kernel rather than a dimension lower bound. Slofstra–Vidick’s negative-eigenspace compression for an approximate central involution is the local predecessor of Lemma 6.5 [SV, proof of Proposition 2.7]. Their Hilbert–Schmidt argument needs quantitative negative-eigenspace dimension. In the operator-norm setting, Bachner–Dogon–Lubotzky subsequently carried out the dimension-changing corner pipeline explicitly. Their Lemmas 2.2–2.3 round an approximate involution and polar-correct an almost-unitary compression, Proposition 2.4 restricts an asymptotic representation to a nonzero almost-invariant corner, and Proposition 1.5 applies these steps to a central Deligne involution before invoking operator–Hilbert–Schmidt stability [BDL, Lemmas 2.2–2.3

and Propositions 1.5 and 2.4]. Thus changing the matrix size to the negative eigenspace, without a positive relative-rank hypothesis, is prior art. Our argument uses the same local corner technology but replaces operator–Hilbert–Schmidt stability and Mal’cev’s theorem by Kazhdan adjoint pinning and stable finiteness, producing an unconditional universal marked kernel.

There is an older analytic ancestor as well. At the foundational level, the conjugation action on Hilbert–Schmidt operators is the $\pi \otimes \bar{\pi}$ model central to Bekka’s theory of amenable unitary representations [Be]; Bekka–Valette connect this model with property (T) and the appearance of finite-dimensional subrepresentations [BV, Theorem 1]. More directly for the operator-norm approximation setting, Dadarlat’s Lemma 3.18 shows that operator-norm almost-commuting finite-rank projections force weak containment of the trivial representation in $\pi \otimes \bar{\pi}$; its proof turns the normalized projections into almost invariant normalized Hilbert–Schmidt vectors for the conjugation representation [Dad, Lemma 3.18 and its proof]. Proposition 3.19, labeled Ozawa–Thom and described there as essentially pointed out by Thom on MathOverflow, then uses property (T) to derive an infinite residually finite quotient [Dad, Proposition 3.19]. These results do not produce a universal marked kernel, but they are direct prior art for the operator-norm-to-Hilbert–Schmidt conjugation maneuver used here.

The broad exact-versus-asymptotic marked-element phenomenon is also older. Slofstra defines the quotients G^{fin} and G^{fa} by elements invisible, respectively, to finite-dimensional representations and to finite-dimensional approximate representations [Slo17, Definitions 2.5–2.7]. This is the natural precedent for the informal “MF radical” language below and for the philosophy of Theorem C. Theorem C is not a claim to originate that separation; it calibrates the same marked compression presentation used in Theorem A in the operator-norm/MF setting. Slofstra–Vidick also recall Fritz’s earlier marked BS(2, 3) hyperlinear-profile example [Fri], restated as [SV, Example 3.1], providing further Baumslag–Solitar context for the cyclic comparison.

De Chiffre–Glebsky–Lubotzky–Thom combine cohomological stability with non-residually-finite central extensions to obstruct Frobenius approximation [DGLT, Introduction]. Their norm, rigidity input, and conclusion differ from ours, but their strategy is an important predecessor for the broader architecture “designated central obstruction plus rigidity of approximate representations.” At the operator/Hilbert–Schmidt boundary, Bachner–Dogon–Lubotzky use a finite normal subgroup as a bridge between the two approximation notions: if $N \trianglelefteq \Lambda$ is finite, Λ/N is hyperlinear, and Λ is MF, then Λ is hyperlinear [BDL, Proposition 1.6]. That result does not kill N ; its proof repairs the restriction to N and passes to a nonzero character-isotypic corner before applying Proposition 2.4 [BDL, proof of Proposition 1.6]. This is direct local prior art for the finite-normal corner. Our argument combines the corner with the one-sided Kazhdan compressor and stable finiteness to force universal annihilation of the finite subgroup, without assuming hyperlinearity

of the quotient. Reformulations of (weak) soficity through equations over groups and extensions involving residually finite groups, locating potential approximation obstructions in finite (central) extensions, appear in Glebsky–Rivera and Glebsky [GR, Gl]. The obstruction of Theorem A likewise lives in a finite central extension, though here the quotient $E/\langle w \rangle$ is not claimed to be residually finite.

The Kazhdan-projection compression observation needed here is proved directly in Lemmas 7.10 and 7.11; we do not attribute that observation to Carrión–Dadarlat–Eckhardt. Their work is used for the group-MF framework and for comparison examples. In particular, their treatment of Abels’ finitely presented solvable group already supplies an amenable MF group that is not residually finite [CDE, Section 2.14 and Corollary 2.18]. Since a finitely generated maximally almost periodic group is residually finite by Mal’cev’s theorem, the broad phenomenon “finitely presented amenable MF but non-MAP” is therefore not new. Theorem C instead gives a calibrated statement about the same marked-sign compression pattern used in Theorem A.

Finally, non-MF-ness of one group C^* -algebra alone does not obstruct the *group* from embedding into a corona, since group MF-ness only requires an MF quotient of $C_{\max}^*(E)$ in which the canonical image of E remains faithful. The contribution asserted here is narrower: stable-finite Kazhdan fixed-space rigidity in a norm matrix corona, together with the central-corner/marked-word device producing a specific nontrivial element in the kernel of every such representation. For context on the approximation landscape see [DGLT, BDL, OAI, KT, Sh, Dad, Slo17, Slo18, SV].

Organization. Section 2 fixes notation and states the external inputs. Section 3 constructs E . Section 4 proves $w \neq 1$ and that Γ embeds in E . Section 5 proves Theorem B. Section 6 develops the corona lemmas. Section 7 defines marked Kazhdan patterns and their compression-defect subgroups, states the finite-normal obstruction criterion (Theorem 7.2), and develops the conjugation representation; Section 8 proves the criterion and deduces Theorem A. Section 9 derives Theorem D and the further consequences. Section 10 proves Theorem C. Section 11 discusses limitations.

2. NOTATION AND EXTERNAL INPUTS

All groups are discrete and countable. All C^* -homomorphisms and C^* -algebra embeddings are understood to be unital unless explicitly stated otherwise. For a unital C^* -algebra A we write $\mathcal{U}(A)$ for its unitary group. On $M_r(\mathbb{C})$ we use the normalized trace tr_r , the operator norm $\|\cdot\|$, and the normalized Hilbert–Schmidt norm $\|x\|_2 = \mathrm{tr}_r(x^*x)^{1/2}$. We will use three elementary facts without further comment: $\|x\|_2 \leq \|x\|$; $|\mathrm{tr}_r(x)| \leq \|x\|$; and $\|uxv\|_2 = \|x\|_2$ for unitaries u, v .

Given a sequence $(d_n)_{n \in J}$ of positive integers indexed by an infinite set $J \subseteq \mathbb{N}$, we write

$$\mathcal{Q}((d_n)_{n \in J}) = \prod_{n \in J}^{\ell^\infty} M_{d_n}(\mathbb{C}) / \bigoplus_{n \in J}^{c_0} M_{d_n}(\mathbb{C}),$$

the quotient of the bounded C^* -product by the ideal of sequences with $\|x_n\| \rightarrow 0$. We call any such algebra a *norm matrix corona*. A *corona representation* of a group E is a group homomorphism $E \rightarrow \mathcal{U}(\mathcal{Q})$ for some norm matrix corona $\mathcal{Q}$; the group E is *MF* if it admits an injective corona representation.

For a group H , $C_{\max}^*(H)$ denotes the maximal group C^* -algebra and $u_h \in C_{\max}^*(H)$ the canonical unitaries. Every unitary representation π of H on a Hilbert space extends uniquely to a $*$ -representation of $C_{\max}^*(H)$, still denoted π . Every group homomorphism $\varphi: H_1 \rightarrow H_2$ induces a unital $*$ -homomorphism $C_{\max}^*(H_1) \rightarrow C_{\max}^*(H_2)$, $u_h \mapsto u_{\varphi(h)}$.

We use the commutator convention $[x, y] = xyx^{-1}y^{-1}$.

The following two classical facts are the only non-elementary external inputs to the analytic kernel argument of Theorem A.

Fact 2.1 (property (T); [BHV, Example 1.7.4(i)]). $\mathrm{SL}_3(\mathbb{Z})$ and $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ have property (T) [BHV, Example 1.7.4(i), via Corollary 1.4.16 and Theorem 1.7.1]; an explicit-constant proof for $\mathrm{SL}_n(\mathbb{Z})$, $n \geq 3$, is [BHV, Theorem 4.2.5]. The group Γ is finitely presented — not as a consequence of property (T), which does not imply finite presentability [BHV, Section 3.4], but by the explicit presentation (4) below, built on [CRW, Theorem 2].

Fact 2.2 (Kazhdan projection; [AW, Va]). Let Γ be a group with property (T). There is a projection $p_\Gamma \in C_{\max}^*(\Gamma)$ such that for every unitary representation π of Γ on a Hilbert space $\mathcal{H}$, the operator $\pi(p_\Gamma)$ is the orthogonal projection onto the subspace $\mathrm{Fix}(\pi) = \{\xi \in \mathcal{H} : \pi(\gamma)\xi = \xi \text{ for all } \gamma \in \Gamma\}$. Moreover $u_\gamma p_\Gamma = p_\Gamma u_\gamma = p_\Gamma$ for every $\gamma \in \Gamma$.

The identities follow from the defining property: the left identity holds in every representation, and the right identity follows by taking adjoints (or replacing γ by γ^{-1}); $C_{\max}^*(\Gamma)$ has a faithful representation.

The only additional input, used exclusively in Section 10, is:

Fact 2.3 ([TWW, Corollary C]). The reduced group C^* -algebra of a countable amenable group is quasidiagonal.

Fact 2.4 (quasidiagonal embedding; [BK, Theorem 3.2.2], see also [BO, Chapter 7]). Every separable unital quasidiagonal C^* -algebra embeds unitaly into a norm matrix corona. Consequently every countable amenable group is MF.

3. THE GROUP

Throughout, $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ with multiplication $(v, A)(w, B) = (v + Aw, AB)$.

We first record concrete presentation data. By [CRW, Theorem 2], $\mathrm{SL}_3(\mathbb{Z})$ has the presentation

$$(3) \quad \left\langle x, y, z \mid \begin{array}{l} x^3 = y^3 = z^2 = (xz)^3 = (yz)^3 = 1, \\ (x^{-1}zxy)^2 = (y^{-1}zyx)^2 = (xy)^6 = 1 \end{array} \right\rangle,$$

where the generators are realized by the matrices

$$x = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad y = \begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad z = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{pmatrix}.$$

The matrix-labelled presentation is also recorded explicitly in [CLV, Introduction]. In particular, the displayed relations are defining relations, rather than merely relations satisfied by the three matrices. Let $v_i = (e_i, 1)$ for $1 \leq i \leq 3$. Reading off the columns of these matrices gives the following complete finite presentation of the semidirect product:

$$(4) \quad \Gamma = \left\langle v_1, v_2, v_3, x, y, z \mid \begin{array}{l} \text{the eight relations in (3),} \\ [v_1, v_2] = [v_1, v_3] = [v_2, v_3] = 1, \\ xv_1x^{-1} = v_3, \quad xv_2x^{-1} = v_1, \quad xv_3x^{-1} = v_2, \\ yv_1y^{-1} = v_1, \quad yv_2y^{-1} = v_2^{-1}v_3, \quad yv_3y^{-1} = v_1v_2^{-1}, \\ zv_1z^{-1} = v_2v_3^{-1}, \quad zv_2z^{-1} = v_1v_3^{-1}, \quad zv_3z^{-1} = v_3^{-1} \end{array} \right\rangle.$$

Lemma 3.1. *The map $\alpha(v, A) = (2v, A)$ is an injective endomorphism of Γ with image $2\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ of index 8, and $a := (e_1, 1) \notin \alpha(\Gamma)$.*

Proof. We compute

$$\begin{aligned} \alpha((v, A)(w, B)) &= \alpha(v + Aw, AB) = (2v + 2Aw, AB) \\ &= (2v, A)(2w, B) = \alpha(v, A)\alpha(w, B), \end{aligned}$$

so α is a homomorphism, clearly injective, with image $2\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$; the index equals $[\mathbb{Z}^3 : 2\mathbb{Z}^3] = 8$. If $(e_1, 1) = (2v, A)$ then $A = 1$ and $e_1 = 2v$, which is impossible in $\mathbb{Z}^3$. $\square$

Let G denote the ascending HNN extension

$$(5) \quad G = \langle \Gamma, t \mid t\gamma t^{-1} = \alpha(\gamma) \ (\gamma \in \Gamma) \rangle.$$

We will use G only as an auxiliary group receiving no injectivity claims; every statement we need about it is either a presentation identity or is verified inside an explicit linear quotient (Lemma 4.2).

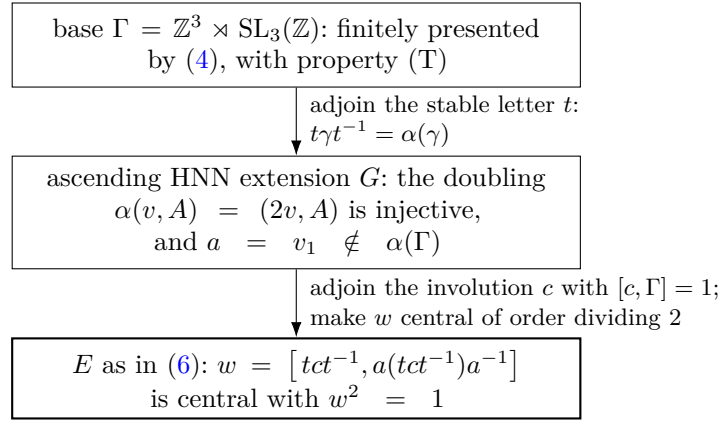


FIGURE 2. Assembly of the group E . The marked word w is nontrivial (Proposition 4.3), yet every finite-dimensional representation over every field kills it (Theorem B), and every norm matrix corona representation kills it (Theorem A).

Definition 3.2. Use the six base generators and the relations displayed in (4), and put $a = v_1$. Define

$$(6) \quad E = \left\langle v_1, v_2, v_3, x, y, z, t, c \left| \begin{array}{l} \text{all relations in (4),} \\ tv_it^{-1} = v_i^2 \quad (1 \leq i \leq 3), \\ txt^{-1} = x, \quad tyt^{-1} = y, \quad tzt^{-1} = z, \\ c^2 = 1, \quad w^2 = 1, \\ [c, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z\}), \\ [w, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z, t, c\}) \end{array} \right. \right\rangle,$$

where w abbreviates the word $[tct^{-1}, v_1(tct^{-1})v_1^{-1}]$.

Thus (6) is an explicit presentation with eight generators and a fully displayed finite list of relators. The presentation identities give a canonical homomorphism $G \rightarrow E$ and a canonical (not a priori injective) homomorphism $\Gamma \rightarrow E$; we write γ, t, c for the images in E . In E the defining relations make w central and impose $w^2 = 1$; Proposition 4.3 below proves that $w \neq 1$. The relations $t\gamma t^{-1} = \alpha(\gamma)$, $[c, \gamma] = 1$ hold for all $\gamma \in \Gamma$ because they hold on generators. Figure 2 summarizes the three-stage assembly and the fate of the marked word.

Remark 3.3 (presentation robustness). Nothing in the proof of Theorem A depends on the particular presentation (4): for *any* finite presentation of Γ , together with words representing the images of the generators under α and a word representing a , the analogue of (6) satisfies Theorem A with the same proof, since the argument consumes the relations only through the canonical homomorphisms $\Gamma \rightarrow E$ and $G \rightarrow E$. The role of [CRW,

Theorem 2] is therefore confined to certifying that (3) is complete as a presentation of $\mathrm{SL}_3(\mathbb{Z})$. The two remaining ingredients of the explicit display are verified independently of [CRW]: the eight relations hold by direct integer computation, and the displayed matrices generate $\mathrm{SL}_3(\mathbb{Z})$ because explicit words in them realize every elementary transvection — for instance $y^{-1}xzy^{-1}x^{-1}$ is the matrix $1 + E_{12}$, and the elementary matrices classically generate $\mathrm{SL}_3(\mathbb{Z})$.

Remark 3.4 (nontriviality of the marked image forces proper infinite compression). Let ρ be any homomorphism of E to a group, and write $\Gamma' = \rho(\Gamma)$. If $\rho(t)\Gamma'\rho(t)^{-1} = \Gamma'$ — in particular whenever Γ' is finite, since a one-sided inclusion of conjugate finite subgroups is an equality — then $a' := \rho(t)^{-1}\rho(a)\rho(t) \in \Gamma'$, so

$$\rho(a(tct^{-1})a^{-1}) = \rho(t)a'\rho(c)a'^{-1}\rho(t)^{-1} = \rho(tct^{-1}),$$

because $\rho(c)$ commutes with Γ' ; hence $\rho(w) = 1$. Consequently any homomorphism with $\rho(w) \neq 1$ compresses Γ' properly, and Γ' is then infinite. This shows structurally why the pattern cannot collide with the standard certificates of MF-ness: an infinite property-(T) image rules out amenable targets, and Theorem B rules out linear and residually finite ones.

Remark 3.5 (why this base). A base for Definition 3.2 must be a finitely presented Kazhdan group with a proper self-embedding. Infinite Kazhdan groups are typically co-Hopfian — lattices in higher-rank semisimple groups by Prasad [Pr], torsion-free hyperbolic groups by Sela [Se] — so such bases are scarce. In $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ the abelian factor exists precisely to break co-Hopfity through the doubling endomorphism, while relative property (T) of the pair $(\Gamma, \mathbb{Z}^3)$ restores property (T) for the whole group.

4. THE WITNESS: $w \neq 1$

Clifford signs, shifts, and HNN doubling have appeared together before in approximate-representation theory, notably in Slofstra's hyperlinear-profile construction [Slo18, Section 2]. The construction below uses the Clifford lamp only as an explicit algebraic witness: the one-sided HNN compression relevant to the obstruction remains in the separate Kazhdan base Γ .

Construction 4.1 (Clifford groups). Let X be a set. Define $\mathrm{Cl}(X)$ to be the quotient of the free unital associative real algebra on symbols $(c_x)_{x \in X}$ by the relations

$$c_x^2 = 1, \quad c_x c_y = -c_y c_x \quad (x \neq y).$$

For completeness, here is a concrete PBW model. Choose a total order on X , and let A_X be the real vector space with basis (e_F) indexed by the finite subsets $F \subseteq X$. Put

$$e_F e_G = (-1)^{\nu(F,G)} e_{F \Delta G}, \quad \nu(F,G) = \#\{(x,y) \in F \times G : y < x\}.$$

Modulo two, ν is additive in each variable with respect to symmetric difference, so this multiplication is associative and $e_\emptyset$ is the identity. The elements $e_{\{x\}}$ square to 1, and for $x \neq y$ exactly one of $\nu(\{x\}, \{y\})$ and $\nu(\{y\}, \{x\})$ is one; hence they anticommute. The defining relations therefore give a homomorphism $\text{Cl}(X) \rightarrow A_X$ sending c_x to $e_{\{x\}}$. Conversely the relations reduce every word to an ordered square-free monomial, whose image is $\pm e_F$. Thus these monomials are a basis, $\text{Cl}(X) \neq 0$, and $-1 \neq 1$ without invoking a normal-form theorem.

Let $\text{ClLamp}(X) \leq \text{Cl}(X)^\times$ be the subgroup generated by $\{\pm 1\} \cup \{c_x : x \in X\}$ and write $z = -1 \in \text{ClLamp}(X)$, a central involution. Every permutation of X induces an algebra automorphism of $\text{Cl}(X)$ mapping $c_x \mapsto c_{\sigma(x)}$ and fixing z ; this gives an action of any group acting on X by automorphisms of $\text{ClLamp}(X)$, and a semidirect product $\text{ClLamp}(X) \rtimes G$ for any G -set X . The group $\text{ClLamp}(X)$ is the central sign extension of $\bigoplus_X \mathbb{Z}/2$ defined by the ordered-inversion cocycle above; its commutator pairing is $(-1)^{|F||G|-|F \cap G|}$. It is a Clifford 2-group of possibly infinite rank, familiar as the multiplicative group generated by Clifford (Majorana) generators in the CAR algebra (see, e.g., [BR2]). The same finite- and infinite-rank Clifford groups with their distinguished central sign appear explicitly in [Slo18, Section 2]; for a modern toolbox of property (T) groups with wreath-like lamp structure see [CIOS].

Lemma 4.2 (linear model). *Embed $\Gamma \hookrightarrow \text{GL}_4(\mathbb{Z})$ by $(v, A) \mapsto \begin{pmatrix} A & v \\ 0 & 1 \end{pmatrix}$ and let $D = \text{diag}(2, 2, 2, 1) \in \text{GL}_4(\mathbb{Z}[1/2])$. Then $D\gamma D^{-1} = \alpha(\gamma)$ for all $\gamma \in \Gamma$, so $\bar{G} := \langle \Gamma, D \rangle \leq \text{GL}_4(\mathbb{Z}[1/2])$ is a quotient of G via $t \mapsto D$. In the $\bar{G}$ -set $\bar{X} := \bar{G}/\Gamma$ the cosets $D\Gamma$ and $aD\Gamma$ are distinct.*

Proof. $D \begin{pmatrix} A & v \\ 0 & 1 \end{pmatrix} D^{-1} = \begin{pmatrix} A & 2v \\ 0 & 1 \end{pmatrix}$, which is the matrix of $\alpha(v, A)$; so the assignment $\gamma \mapsto \gamma$, $t \mapsto D$ respects the relations (5). If $aD\Gamma = D\Gamma$ then $D^{-1}aD \in \Gamma$ inside $\text{GL}_4(\mathbb{Z}[1/2])$. But $D^{-1}aD = \begin{pmatrix} 1 & \frac{1}{2}e_1 \\ 0 & 1 \end{pmatrix}$ has a non-integral entry, so it does not lie in the integral group Γ . $\square$

The three sites used by the witness are summarized in Figure 3. Only the inequality of the two right-hand sites is needed; their Clifford generators then anticommute automatically.

Proposition 4.3. *$w \neq 1$ in E .*

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Proof. Let $\bar{X} = \bar{G}/\Gamma$ as in Lemma 4.2 and form $W := \text{ClLamp}(\bar{X}) \rtimes \bar{G}$ as in Construction 4.1. Define a map on the generators of (6) by

$$\begin{aligned} g &\mapsto g \in \bar{G} \quad (g \in \{v_1, v_2, v_3, x, y, z\}), \\ t &\mapsto D \in \bar{G}, \quad c \mapsto c_o, \quad o := \Gamma \in \bar{X}. \end{aligned}$$

We check the relations. The base relations in (4) and $tgt^{-1} = \alpha(g)$ for the six base generators hold in $\bar{G}$ by Lemma 4.2. We have $c_o^2 = 1$. For $\gamma \in \Gamma$ we

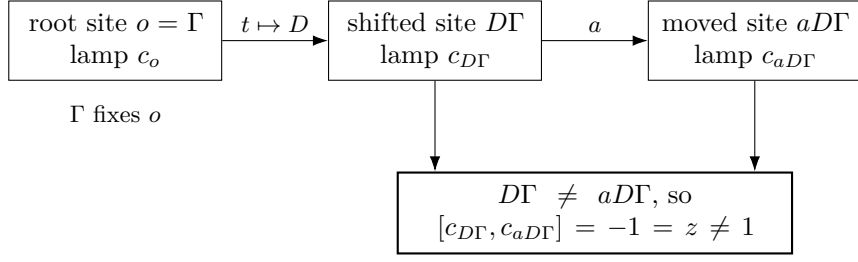


FIGURE 3. The Clifford witness for $w \neq 1$. The base fixes the root lamp, while the proper doubling shift creates two distinct Clifford sites whose commutator is the central sign.

have $\gamma o = o$, so $\gamma c_o \gamma^{-1} = c_{\gamma o} = c_o$, giving the six commutation relations for c . Under this assignment,

$$t c t^{-1} \mapsto c_{D\Gamma}, \quad a(t c t^{-1}) a^{-1} \mapsto c_{aD\Gamma},$$

and the two sites are distinct by Lemma 4.2, so

$$w \mapsto [c_{D\Gamma}, c_{aD\Gamma}] = c_{D\Gamma} c_{aD\Gamma} c_{D\Gamma}^{-1} c_{aD\Gamma}^{-1} = -1 = z.$$

Since z is a central involution of W , the remaining relations $w^2 = 1$ and $[w, s] = [w, t] = [w, c] = 1$ are also satisfied. Hence the assignment extends to a group homomorphism $E \rightarrow W$ sending w to $z \neq 1$, and therefore $w \neq 1$ in E . $\square$

Remark 4.4. The target uses the infinite group $\text{CLamp}(\bar{X})$ when $\bar{X}$ is infinite, but each group element involves only finitely many Clifford sites; the remaining data are matrices over $\mathbb{Z}[1/2]$. No normal-form theory for HNN extensions enters. In particular the consistency of the presentation (6) is verified by an explicit linear-algebraic model.

Corollary 4.5. *The canonical homomorphisms $\Gamma \rightarrow E$ and $\Gamma \rightarrow G$ are injective; in particular E contains an isomorphic copy of Γ .*

Proof. The homomorphism $E \rightarrow W$ constructed in the proof of Proposition 4.3 sends each of the six base generators to the corresponding element of $\bar{G} \leq W$, so the composite $\Gamma \rightarrow E \rightarrow W$ agrees on generators with, hence equals, the faithful linear embedding $\Gamma \hookrightarrow \text{GL}_4(\mathbb{Z}) \leq \bar{G}$ of Lemma 4.2. A homomorphism whose composite with another homomorphism is injective is itself injective; this gives injectivity of $\Gamma \rightarrow E$, and of $\Gamma \rightarrow G$, since the latter composes with $G \rightarrow E$ to the former. $\square$

5. THE FINITE-DIMENSIONAL OBSTRUCTION

We prove Theorem B. Everything happens inside the finite-dimensional algebra $\text{End}_k(V)$.

Proof of Theorem B. Let $\pi: H \rightarrow \text{GL}(V)$ be a homomorphism and let

$$C = \{x \in \text{End}_k(V) : \pi(\gamma)x = x\pi(\gamma) \text{ for all } \gamma \in \Gamma\}$$

be the commutant of $\pi(\Gamma)$, a linear subspace of $\text{End}_k(V)$. Let $u = \pi(t)$ and consider the linear automorphism $\Phi(x) = uxu^{-1}$ of $\text{End}_k(V)$.

We claim $C \subseteq \Phi(C)$. Let $x \in C$ and set $y = u^{-1}xu$, so that $x = \Phi(y)$. For $\gamma \in \Gamma$ we have $t\gamma t^{-1} \in \Gamma$, hence $\pi(t\gamma t^{-1})$ commutes with x , and therefore

$$\pi(\gamma)y = u^{-1}\pi(t\gamma t^{-1})xu = u^{-1}x\pi(t\gamma t^{-1})u = y\pi(\gamma).$$

Thus $y \in C$, proving the claim. Since Φ is a linear isomorphism, $\dim_k \Phi(C) = \dim_k C < \infty$, and the inclusion $C \subseteq \Phi(C)$ forces

$$(7) \quad C = \Phi(C) = uCu^{-1}.$$

Now $\pi(c) \in C$ because c centralizes Γ . By (7), $\pi(tct^{-1}) = u\pi(c)u^{-1} \in C$. Since $a \in \Gamma$, the element $\pi(tct^{-1})$ commutes with $\pi(a)$, so $\pi(a(tct^{-1})a^{-1}) = \pi(tct^{-1})$, and therefore

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = [\pi(tct^{-1}), \pi(tct^{-1})] = 1. \quad \square$$

The same calculation proves the final, subgroup-valued assertion of Theorem B. The elements of H whose conjugation action preserves C in both directions form a subgroup. The argument above puts every one-sided compressor in that subgroup, and therefore puts all of $G_{\text{comp}}(\Gamma)$ in it. Thus, for $g \in G_{\text{comp}}(\Gamma)$ and $z \in C_H(\Gamma)$, the operator $\pi(gzg^{-1})$ still belongs to C and commutes with every $\pi(\ell)$, $\ell \in \Gamma$. Hence π kills all the generators in (2), and normality of its kernel kills their normal closure.

Corollary 5.1. *The group E of Definition 3.2 admits no injective homomorphism into $\text{GL}(V)$ for any finite-dimensional vector space V over any field. More strongly, finite-dimensional linear representations do not separate its points: they all identify w with 1. In particular E is not maximally almost periodic or residually finite, $C_{\text{max}}^*(E)$ is not residually finite-dimensional as a C^* -algebra, and no finite quotient of E can witness $w \neq 1$.*

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Proof. Apply Theorem B with $H = E$ and the images of Γ , t , c , a in E ; the hypotheses hold by the defining relations (6). Every finite-dimensional representation kills w , and $w \neq 1$ by Proposition 4.3. Moreover the canonical unitary u_w is not 1 in $C_{\text{max}}^*(E)$, because the regular representation sends it to $\lambda_w \neq 1$. Every finite-dimensional $*$ -representation of $C_{\text{max}}^*(E)$ restricts to a finite-dimensional representation of E and therefore kills $u_w - 1$; hence $C_{\text{max}}^*(E)$ is not residually finite-dimensional. A finite quotient of E gives a finite-dimensional representation after composition with a faithful permutation representation, so it also kills w . $\square$

Remark 5.2. The proof of Theorem B used only that a finite-dimensional subspace cannot be properly contained in an isomorphic image of itself. The corona of (1) is not finite-dimensional, and the commutant of the image of Γ there is not a coordinatewise object, so the argument does not transfer as stated. Property (T) repairs exactly this: the Kazhdan projection turns “the Γ -fixed subspace” into a single element of the corona, to which the finiteness

of the corona (Lemma 6.3) can be applied. The rest of the paper carries this out.

6. NORM MATRIX CORONAS

Throughout this section $\mathcal{Q} = \mathcal{Q}((d_n)_{n \in J})$ is a norm matrix corona, $q: \prod_{n \in J} M_{d_n}(\mathbb{C}) \rightarrow \mathcal{Q}$ the quotient map.

The following lifting facts are standard perturbation theory; see Loring [Lo] for a systematic treatment.

Lemma 6.1 (lifting).

- (1) Every projection $e \in \mathcal{Q}$ lifts to a sequence $(e_n)_{n \in J}$ of projections, $e_n \in M_{d_n}(\mathbb{C})$.
- (2) Every unitary $u \in \mathcal{Q}$ lifts to a sequence of unitaries.

Proof. (1) Lift e to a bounded sequence (x_n) ; replacing x_n by $\frac{1}{2}(x_n + x_n^*)$ and then by the functional-calculus cutoff $f(x_n)$, where $f(t) = \min(1, \max(0, t))$, we may assume $0 \leq x_n \leq 1$. These replacements preserve the quotient class because functional calculus commutes with the quotient map and $f(e) = e$. Since $q((x_n))$ is a projection, $\|x_n^2 - x_n\| \rightarrow 0$, so the spectrum of x_n is eventually contained in $[0, \frac{1}{3}] \cup (\frac{2}{3}, 1]$; for such n put $e_n = \chi_{(\frac{1}{2}, \infty)}(x_n)$, a projection with $\|e_n - x_n\| \rightarrow 0$, and set $e_n = 0$ for the finitely many remaining n . Then (e_n) lifts e .

(2) Lift u to an arbitrary bounded sequence (x_n) . Since $q((x_n))$ is unitary, $\|x_n^* x_n - 1\| \rightarrow 0$, so x_n is eventually invertible; for such n put $u_n = x_n(x_n^* x_n)^{-1/2}$, a unitary with $\|u_n - x_n\| \rightarrow 0$; set $u_n = 1$ for the remaining n . $\square$

The following elementary amplification trick normalizes separation constants; it is used in Section 9.

Lemma 6.2 (separation amplifier). *Let $u, v \in \mathcal{U}(M_r(\mathbb{C}))$ with $\|u - v\| = \varepsilon > 0$. Then there is an integer $1 \leq n \leq \lceil \pi/(3\varepsilon) \rceil$ with*

$$\|u^{\otimes n} - v^{\otimes n}\| \geq 1.$$

Moreover $\|a^{\otimes n} - b^{\otimes n}\| \leq n \|a - b\|$ for all contractions $a, b \in M_r(\mathbb{C})$ and all $n \geq 1$.

Proof. Write $w = u^* v$, so that $\|u - v\| = \|1 - w\|$, and choose an eigenvalue $e^{i\theta}$ of the unitary w with $|1 - e^{i\theta}| = \varepsilon$ and $0 < |\theta| \leq \pi$; then $\varepsilon = 2 \sin(|\theta|/2) \leq |\theta|$. Let $n = \lceil \pi/(3|\theta|) \rceil$, the least integer with $n|\theta| \geq \pi/3$. If $n = 1$ then $n|\theta| = |\theta| \leq \pi$; otherwise $|\theta| < \pi/3$ and minimality gives $n|\theta| < \pi/3 + |\theta| < 2\pi/3$. In both cases $\pi/3 \leq n|\theta| \leq \pi$, so $|1 - e^{in\theta}| = 2 \sin(n|\theta|/2) \geq 2 \sin(\pi/6) = 1$. Since $e^{in\theta}$ is an eigenvalue of $w^{\otimes n}$ and $u^{\otimes n} - v^{\otimes n} = u^{\otimes n}(1 - w^{\otimes n})$,

$$\|u^{\otimes n} - v^{\otimes n}\| = \|1 - w^{\otimes n}\| \geq |1 - e^{in\theta}| \geq 1,$$

and $n \leq \lceil \pi/(3\varepsilon) \rceil$. The last claim is the telescoping identity $a^{\otimes n} - b^{\otimes n} = \sum_{j=0}^{n-1} b^{\otimes j} \otimes (a - b) \otimes a^{\otimes (n-1-j)}$ together with submultiplicativity of the operator norm under tensor products of contractions. $\square$

Stable finiteness of norm matrix coronas is standard and goes back to Blackadar–Kirchberg [BK]; we include the short proofs to keep the external surface explicit.

Lemma 6.3 (stable finiteness). *Let $\mathcal{Q}$ be a norm matrix corona.*

- (1) *Every isometry in $\mathcal{Q}$ is a unitary; the same holds in every matrix amplification $M_k(\mathcal{Q})$.*
- (2) *In any unital C^* -algebra in which every isometry is a unitary — by (1), in $\mathcal{Q}$ and its matrix amplifications — if $p \leq q$ are projections with $v^*v = q$ and $vv^* = p$ for some element v , then $p = q$.*

Proof. (1) Let $v \in \mathcal{Q}$ with $v^*v = 1$ and lift v to (x_n) with $\|x_n^*x_n - 1\| \rightarrow 0$. Eventually x_n is invertible (a left-invertible square matrix is invertible), and the unitary $u_n = x_n(x_n^*x_n)^{-1/2}$ satisfies $\|u_n - x_n\| \rightarrow 0$; hence $v = q((u_n))$ is unitary. For $M_k(\mathcal{Q})$ note $M_k(\mathcal{Q}) \cong \mathcal{Q}((k d_n)_{n \in J})$, since matrix amplification commutes with products and with the ideal of null sequences.

(2) This is the standard fact that in a finite algebra a projection is not equivalent to a proper subprojection; we include the short verification for completeness. As in any C^* -algebra, $v^*v = q$ gives $v = vq$ (because $\|v(1-q)\|^2 = \|(1-q)v^*v(1-q)\| = \|(1-q)q(1-q)\| = 0$) and $vv^* = p$ gives $v = pv$ (because $\|(1-p)v\|^2 = \|(1-p)vv^*(1-p)\| = 0$). From $p \leq q$ we get $qv = q(pv) = pv = v$, hence $(1-q)v = 0$ and, taking adjoints, $v^*(1-q) = 0$ and $(1-q)v^* = (v(1-q))^* = 0$. Then $s := v + (1-q)$ satisfies

$$s^*s = v^*v + v^*(1-q) + (1-q)v + (1-q) = q + 0 + 0 + (1-q) = 1,$$

so s is an isometry, and

$$ss^* = vv^* + v(1-q) + (1-q)v^* + (1-q) = p + 0 + 0 + (1-q) = p + (1-q).$$

By finiteness of the ambient algebra, $ss^* = 1$, i.e. $p + (1-q) = 1$, i.e. $p = q$. $\square$

Lemma 6.4 (corner coronas). *Let $e \in \mathcal{Q}$ be a nonzero projection. Then there are an infinite set $J' \subseteq J$ and positive integers $(r_n)_{n \in J'}$ with $r_n \leq d_n$ such that $e\mathcal{Q}e$ is unital $*$ -isomorphic to the norm matrix corona $\mathcal{Q}' = \mathcal{Q}((r_n)_{n \in J'})$, by an isomorphism carrying e to 1.*

Proof. By Lemma 6.1(1) lift e to projections (e_n) ; let $J' = \{n \in J : e_n \neq 0\}$ and $r_n = \text{rank } e_n$ for $n \in J'$. The set J' is infinite, since otherwise (e_n) would be a null sequence and $e = 0$.

Consider the map

$$\Psi: \prod_{n \in J'} e_n M_{d_n}(\mathbb{C}) e_n \longrightarrow e\mathcal{Q}e, \quad (x_n)_{n \in J'} \longmapsto q((\tilde{x}_n)_{n \in J}),$$

where $\tilde{x}_n = x_n$ for $n \in J'$ and $\tilde{x}_n = 0$ otherwise. Ψ is a $*$ -homomorphism onto $e\mathcal{Q}e$: surjectivity holds because any $y \in e\mathcal{Q}e$ lifts to (y_n) with $\|e_n y_n e_n - y_n\| \rightarrow 0$, and $e_n y_n e_n = 0$ off J' . Its kernel is exactly the

sequences with $\|x_n\| \rightarrow 0$. Choosing unitary identifications $e_n \mathbb{C}^{d_n} \cong \mathbb{C}^{r_n}$ identifies $e_n M_{d_n}(\mathbb{C}) e_n$ with $M_{r_n}(\mathbb{C})$ compatibly with adjoints and norms, so

$$e \mathcal{Q} e \cong \mathcal{Q}((r_n)_{n \in J'}) = \mathcal{Q}',$$

a unital isomorphism carrying e to 1. $\square$

Lemma 6.5 (finite-normal corners). *Let H be a group, $F \trianglelefteq H$ a finite normal subgroup, and $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ a corona representation whose restriction to F is nontrivial. Then there are an infinite set $J' \subseteq J$, positive integers $(r_n)_{n \in J'}$ with $r_n \leq d_n$, and a corona representation*

$$\Theta_-: H \rightarrow \mathcal{U}(\mathcal{Q}'), \quad \mathcal{Q}' = \mathcal{Q}((r_n)_{n \in J'}),$$

with

$$\sum_{f \in F} \Theta_-(f) = 0.$$

In the case $F = \{1, w\}$ with w a central involution and $\Theta(w) \neq 1$, the corner projection is $e = \frac{1}{2}(1 - \Theta(w))$ and the conclusion reads $\Theta_-(w) = -1$.

In the involution case, this is the norm-corona counterpart of Slofstra–Vidick’s compression to the (-1) -eigenspace of a repaired approximate central involution [SV, proof of Proposition 2.7]. In their normalized Hilbert–Schmidt setting a quantitative separation supplies a positive relative dimension. Here nonzeroness of the corona projection is enough: passing to its corner changes the ambient matrix sequence and renormalizes all later traces there. The operator-norm involution version of this corner-and-polar-correction pipeline appears in [BDL, Proposition 1.5], with rounding and compression inputs in [BDL, Lemma 2.2 and Proposition 2.4]; the character-isotypic finite-normal variant appears in the proof of [BDL, Proposition 1.6]. For general F the corner projection is the complement of the standard finite-group Reynolds projection of the finite normal subgroup.

Proof. Put

$$p_F = \frac{1}{|F|} \sum_{f \in F} \Theta(f).$$

Then $p_F^* = p_F$, because $\Theta(F)$ is closed under adjoints (F is closed under inverses), and $p_F^2 = p_F$: for each $f_0 \in F$, left translation by f_0 permutes F , so $\Theta(f_0) p_F = p_F$, and averaging over f_0 gives $p_F^2 = p_F$. Thus p_F is a projection. For $g \in H$, conjugation by g permutes the normal subgroup F , so $\Theta(g) p_F \Theta(g)^{-1} = p_F$: the projection p_F commutes with $\Theta(H)$. If $p_F = 1$ then, for every $f_0 \in F$, $\Theta(f_0) = \Theta(f_0) p_F = p_F = 1$, contradicting nontriviality of $\Theta|_F$; so $e := 1 - p_F$ is a nonzero projection commuting with $\Theta(H)$.

By Lemma 6.4, $e \mathcal{Q} e \cong \mathcal{Q}'$ by a unital $*$ -isomorphism carrying e to 1. Since e commutes with $\Theta(H)$, the map $\Theta_-(g) := \Theta(g)e$ is a unital group

homomorphism $H \rightarrow \mathcal{U}(e\mathcal{Q}e) = \mathcal{U}(\mathcal{Q}')$, and

$$\sum_{f \in F} \Theta_-(f) = \left(\sum_{f \in F} \Theta(f) \right) e = |F| p_F (1 - p_F) = 0.$$

For $F = \{1, w\}$ with w central of order two, $p_F = \frac{1}{2}(1 + \Theta(w))$, so $e = \frac{1}{2}(1 - \Theta(w))$ and $\Theta_-(w) = \Theta(w)e = (1 - 2e)e = -e = -1$ in $\mathcal{Q}'$. $\square$

7. THE CONJUGATION REPRESENTATION AND THE KAZHDAN PROJECTION

The corona half of Theorem A will be deduced from a general criterion, whose data we now fix.

Definition 7.1 (marked Kazhdan pattern). Let H be a countable group. A *marked Kazhdan pattern* in H is a tuple (Λ, ι, t, c) , where Λ is a countable group with property (T), $\iota: \Lambda \rightarrow H$ is a group homomorphism — *not* assumed injective — and $t, c \in H$, subject to

- (M1) $t \iota(\lambda) t^{-1} \in \iota(\Lambda)$ for every $\lambda \in \Lambda$;
- (M2) $c \iota(\lambda) = \iota(\lambda) c$ for every $\lambda \in \Lambda$.

Write $d := tct^{-1}$. The *compression-defect subgroup* of the pattern is the normal closure

$$N_{\text{comp}} = \langle\langle [d, \iota(\lambda)] : \lambda \in \Lambda \rangle\rangle_H \trianglelefteq H.$$

There are two canonical enlargements. The integer-orbit defect is

$$N_{\text{orb}} = \langle\langle [t^n c t^{-n}, \iota(\lambda)] : n \in \mathbb{Z}, \lambda \in \Lambda \rangle\rangle_H,$$

and, for $L = \iota(\Lambda)$, the intrinsic defect is $\mathfrak{D}(H, L)$ from (2). Since t is a one-sided compressor and $G_{\text{comp}}(L)$ is a group, all powers t^n , including negative powers, lie in $G_{\text{comp}}(L)$. Hence

$$(8) \quad N_{\text{comp}} \leq N_{\text{orb}} \leq \mathfrak{D}(H, L).$$

Theorem 7.2 (finite-normal obstruction criterion). *Let H be a countable group with a marked Kazhdan pattern (Λ, ι, t, c) , and let $F \trianglelefteq H$ be a finite normal subgroup with $F \subseteq N_{\text{comp}}$. Then every corona representation $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ satisfies $\Theta(f) = 1$ for all $f \in F$.*

Theorem 7.3 (normal-Kazhdan radical criterion). *Let H be countable with a marked Kazhdan pattern, and let $K \trianglelefteq H$ have property (T). If $K \leq N_{\text{comp}}$, then*

$$K \leq \text{Rad}_{\text{MF}}(H).$$

Thus every norm-matrix-corona representation of H kills K . In particular, if $K \neq 1$, then H is non-MF. The obstruction subgroup need not be finite, central, or torsion.

Proof. Fix a homomorphism Θ from H to an operator-norm matrix ultra-product, and suppose that $\Theta(k) \neq 1$ for some $k \in K$. Replace the target by the countable range $R = \Theta(H)$. The subgroup $K' = \Theta(K) \leq R$ is nontrivial

and normal, has property (T), and is contained in the compression defect of the transported pattern.

The inclusion of R into the original ultraproduct is injective, so the universal MF radical of R is trivial. The universal-quotient diagonalization therefore gives a faithful representation of R in one cofinite norm matrix corona. Finally fix a finite test set $F \subseteq R$. Each ordered pair of distinct elements of F is separated, by faithfulness, in infinitely many coordinates of the corona; Lemma 6.2 assigns to the pair a tensor exponent — different pairs need different exponents — which raises that pair's separation to 1 while multiplying the multiplicative defect by a bounded factor. Because the pairs' infinite sets of good coordinates need not intersect, the amplified coordinates are combined by a block sum, one block per ordered pair; the operator norm of a block sum is the maximum over blocks, so the assembled unitaries separate all pairs of F by 1 at once. Hence R has a uniformly separated operator-norm matrix approximation.

We now cut by a second Kazhdan projection. Let $p_{K'}$ be the Kazhdan projection of K' and let $\tilde{\Theta}$ denote the uniformly separated matrix model. For a surjective endomorphism θ of K' and any representation π , the projection $\pi(\theta_*(p_{K'})) =$ the projection onto the $\pi(\theta(K'))$ -fixed vectors is again the projection onto the $\pi(K')$ -fixed vectors; applying this to the conjugation automorphisms of the normal subgroup K' shows that $e := 1 - \tilde{\Theta}(p_{K'})$ commutes with the image of $\tilde{\Theta}$, so cutting by e is again a homomorphism of H . The corner is nonzero: if $e = 0$ then, by the identity $u_k p_{K'} = p_{K'}$, every $k \in K'$ satisfies $\tilde{\Theta}(k) = \tilde{\Theta}(k)\tilde{\Theta}(p_{K'}) = \tilde{\Theta}(p_{K'}) = 1$, contradicting faithfulness on the nontrivial K' .

On the corner, the restriction to K' has no invariant vectors, so by property (T) it has no almost-invariant vectors either: fix a finite Kazhdan set $Q_0 \subseteq K'$ with constant $\kappa > 0$. If a corner coordinate of size r satisfied $\|U_k - 1\|_2 \leq \varepsilon$ for all $k \in Q_0$, then averaging over an orthonormal basis (e_i) ,

$$\frac{1}{r} \sum_{i=1}^r \sum_{k \in Q_0} \|(U_k - 1)e_i\|^2 \leq |Q_0| \varepsilon^2,$$

so some e_i is $(Q_0, \sqrt{|Q_0|} \varepsilon)$ -almost invariant; for $\varepsilon < \kappa/\sqrt{|Q_0|}$ this is impossible. Hence along the corner coordinates $\max_{k \in Q_0} \|U_k - 1\|_2 \geq \kappa/\sqrt{|Q_0|}$: the set Q_0 survives tracially. On the other hand, the compressor argument of Steps 1–3 of Section 8, run in this corner model, sends $\|\cdot - 1\|_2 \rightarrow 0$ for every generator $[tct^{-1}, \iota(\lambda)]$ of the transported defect, and the telescoping of Step 4 propagates the collapse to every element of the subgroup K' of its normal closure — in particular to the finite set Q_0 — a contradiction. This second Kazhdan projection replaces the finite Reynolds average of Theorem 7.2. $\square$

Corollary 7.4 (Kazhdan part and exact radical). *Let $\mathcal{D}_T$ be the subgroup generated by all normal property (T) subgroups of H contained in N_{comp} . Then $\mathcal{D}_T \leq \text{Rad}_{\text{MF}}(H)$. If N_{comp} itself has property (T), every corona representation factors uniquely through H/N_{comp} ; and if this quotient is MF,*

then

$$\text{Rad}_{\text{MF}}(H) = N_{\text{comp}}.$$

Theorem 7.5 (intrinsic normal-Kazhdan criterion). *Let $L = \iota(\Lambda) \leq H$ and let $\mathfrak{D}(H, L)$ be the compression-centralizer defect from (2). If $K \trianglelefteq H$ has property (T) and $K \leq \mathfrak{D}(H, L)$, then*

$$K \leq \text{Rad}_{\text{MF}}(H).$$

Consequently, if $\mathfrak{D}(H, L)$ itself has property (T), every corona representation factors through $H/\mathfrak{D}(H, L)$; if that quotient is MF, then

$$\text{Rad}_{\text{MF}}(H) = \mathfrak{D}(H, L).$$

The same containment holds for the subgroup generated by all normal property (T) subgroups contained in $\mathfrak{D}(H, L)$.

Proof. In the conjugation representation associated to a corona model, let P be the Kazhdan projection onto the L -fixed vectors. For every one-sided compressor $s \in \mathcal{C}(L)$ the fixed-vector calculation gives $P \leq \beta(s)P\beta(s)^*$. Stable finiteness upgrades this to equality, so P commutes with $\beta(s)$. It therefore commutes with $\beta(g)$ for every $g \in G_{\text{comp}}(L)$, including inverses and arbitrary finite products.

If $z \in C_H(L)$, then the vector represented by $\Theta(z)$ is L -fixed. Hence the vector represented by $\Theta(gzg^{-1})$ is again L -fixed for every $g \in G_{\text{comp}}(L)$. It follows that every generator $[gzg^{-1}, \ell]$ of (2), and therefore its normal closure, is trivial in the associated tracial matrix ultraproduct. The second Kazhdan-corner argument from Theorem 7.3, now with $K \leq \mathfrak{D}(H, L)$, gives the radical containment. The factorization and equality statements follow from the universal property of the quotient and Proposition 9.3. $\square$

Theorem 7.6 (faithfully tracial targets). *Let (Λ, ι, t, c) be a marked Kazhdan pattern in the countable group H and $L = \iota(\Lambda)$, as in Definition 7.1. Let A be a unital C^* -algebra carrying a faithful tracial state τ , and suppose that $A \otimes_{\max} A^{\text{op}}$ is stably finite. For every homomorphism $\Theta: H \rightarrow \mathcal{U}(A)$,*

$$\Theta(\mathfrak{D}(H, L)) = 1.$$

In particular the conclusion holds when A is nuclear and faithfully tracial.

Proof. Consider the exact representation

$$\beta(g) = \Theta(g) \otimes \Theta(g)^{*, \text{op}} \quad \text{in } \mathcal{U}(A \otimes_{\max} A^{\text{op}}).$$

On $L^2(A, \tau)$ its left-right action is conjugation by $\Theta(g)$. The Kazhdan projection for L therefore projects onto the fixed vectors of this conjugation action. The one-sided compressor calculation gives the same projection order as above, and stable finiteness of the maximal tensor product turns it into equality. Thus the fixed space is invariant under $G_{\text{comp}}(L)$. Since $\Theta(z)$ is fixed whenever $z \in C_H(L)$, so is $\Theta(gzg^{-1})$ for every $g \in G_{\text{comp}}(L)$. It commutes in $L^2(A, \tau)$ with every $\Theta(\ell)$, $\ell \in L$; faithfulness of τ promotes this

L^2 equality to equality in A . Hence all generators of $\mathfrak{D}(H, L)$ vanish under Θ .

If A is nuclear, maximal and minimal tensor products agree. The product trace on $A \otimes_{\min} A^{\text{op}}$ is faithful, so this tensor product is stably finite. $\square$

Remark 7.7 (higher Kazhdan sectors). The preceding arguments use the isolated trivial representation of Λ . A plausible higher-sector extension replaces it by a finite-dimensional representation ρ with $\rho \circ \alpha \simeq \rho$, where α is the compression endomorphism. A tuple of lamp vectors transforming as ρ should be pinned by the corresponding isotypic Kazhdan projection, producing vanishing covariance defects rather than commutators. Brown's finite-dimensional-sector Kazhdan projections [Br] suggest this formulation, but we do not claim the extension as a theorem here: the covariance data and the passage from a one-sided sector inclusion to its precise defect relations still have to be written out.

Remark 7.8. Theorem A(2) is the instance $H = E$, $\Lambda = \Gamma$, ι the canonical homomorphism, and $F = \{1, w\}$; the membership $w \in N_{\text{comp}}$ is the identity $w = [d, \iota(a)]^2$ of Lemma 8.1 below. The property (T) hypothesis cannot simply be deleted: Theorem C realizes the pattern over the base $\mathbb{Z}$ in an MF group in which the marked involution survives. As with Theorem A, this is not a claim that every possible proof must proceed through property (T); a substitute furnishing the uniform fixed-space projection is what the argument consumes.

For the remainder of this section and in Section 8, fix a countable group H with a marked Kazhdan pattern (Λ, ι, t, c) and a finite normal subgroup $F \trianglelefteq H$ with $F \subseteq N_{\text{comp}}$, and suppose given a corona representation $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ whose restriction to F is nontrivial; the proof of Theorem 7.2 will derive a contradiction. By Lemma 6.5 there is a corona representation $\Theta_-: H \rightarrow \mathcal{U}(\mathcal{Q}')$, $\mathcal{Q}' = \mathcal{Q}((r_n)_{n \in J'})$, with $\sum_{f \in F} \Theta_-(f) = 0$. Using Lemma 6.1(2), choose for every $g \in H$ a sequence of unitaries $U_{g,n} \in M_{r_n}(\mathbb{C})$ lifting $\Theta_-(g)$, with $U_{1,n} = 1$.

Let $K_n = L^2(M_{r_n}(\mathbb{C}), \text{tr}_{r_n})$, a Hilbert space of dimension r_n^2 whose bounded operators $B(K_n)$ form the matrix algebra $M_{r_n^2}(\mathbb{C})$, and let

$$\mathcal{B} = \prod_{n \in J'} B(K_n) / \bigoplus_{n \in J'} B(K_n),$$

again a norm matrix corona. For a unitary $U \in M_{r_n}(\mathbb{C})$ let $\text{Ad } U \in \mathcal{U}(B(K_n))$ denote $x \mapsto UxU^*$.

Lemma 7.9.

(1) For unitaries $U, V \in M_r(\mathbb{C})$ and every $x \in M_r(\mathbb{C})$,

$$\|UxU^* - VxV^*\|_2 \leq 2 \|U - V\| \|x\|_2,$$

hence $\|\text{Ad } U - \text{Ad } V\| \leq 2 \|U - V\|$ as operators on the Hilbert space K_n (and therefore as elements of $B(K_n)$).

- (2) The class $\beta(g)$ of the sequence $(\text{Ad } U_{g,n})_{n \in J'}$ in $\mathcal{B}$ defines a group homomorphism $\beta: H \rightarrow \mathcal{U}(\mathcal{B})$, independent of the choice of unitary lifts.
- (3) β extends to a unital $*$ -homomorphism $\tilde{\beta}: C_{\max}^*(H) \rightarrow \mathcal{B}$ with $\tilde{\beta}(u_g) = \beta(g)$.

Proof. (1) Since $UxU^* - VxV^* = (U - V)xU^* + Vx(U - V)^*$, the ideal property of the normalized Hilbert–Schmidt norm gives

$$\begin{aligned} \|UxU^* - VxV^*\|_2 &\leq \|(U - V)xU^*\|_2 + \|Vx(U - V)^*\|_2 \\ &\leq 2\|U - V\| \|x\|_2. \end{aligned}$$

Taking the supremum over $\|x\|_2 = 1$ proves the asserted operator-norm bound on K_n .

(2) If $(U_{g,n})$ and $(U'_{g,n})$ both lift $\Theta_-(g)$ then $\|U_{g,n} - U'_{g,n}\| \rightarrow 0$, so by (1) the two sequences of adjoint maps agree in $\mathcal{B}$; this is independence of lifts. For multiplicativity, $U_{g,n}U_{h,n}$ and $U_{gh,n}$ both lift $\Theta_-(gh)$, so $\|U_{g,n}U_{h,n} - U_{gh,n}\| \rightarrow 0$ and, by (1) again, $\beta(g)\beta(h) = \beta(gh)$. Finally $\beta(1) = 1$ since $U_{1,n} = 1$, so each $\beta(g)$ is a unitary with inverse $\beta(g^{-1})$.

(3) This is precisely the universal property of the maximal group C^* -algebra: the homomorphism $\beta: H \rightarrow \mathcal{U}(\mathcal{B})$ integrates uniquely to a unital $*$ -homomorphism $\tilde{\beta}: C_{\max}^*(H) \rightarrow \mathcal{B}$ satisfying $\tilde{\beta}(u_g) = \beta(g)$. $\square$

Let $\iota_*: C_{\max}^*(\Lambda) \rightarrow C_{\max}^*(H)$ be the unital $*$ -homomorphism induced by ι (Section 2), and let $p_\Lambda \in C_{\max}^*(\Lambda)$ be the Kazhdan projection of Fact 2.2. Composing, $\tilde{\beta} \circ \iota_*$ is a unital $*$ -homomorphism $C_{\max}^*(\Lambda) \rightarrow \mathcal{B}$, and we write

$$P := \tilde{\beta}(\iota_*(p_\Lambda)) \in \mathcal{B},$$

a projection.

Lemma 7.10 (order identity). *Let H_1 be a countable group, Λ_1 a countable group with property (T), $\iota_1: \Lambda_1 \rightarrow H_1$ a homomorphism, and $t_1 \in H_1$ an element with $t_1 \iota_1(\lambda) t_1^{-1} \in \iota_1(\Lambda_1)$ for every $\lambda \in \Lambda_1$; write $\bar{p} = (\iota_1)_*(p_{\Lambda_1}) \in C_{\max}^*(H_1)$. Then, in $C_{\max}^*(H_1)$:*

- (1) $u_{\iota_1(\lambda)} \bar{p} = \bar{p}$ for all $\lambda \in \Lambda_1$;
- (2) $\bar{p} (u_{t_1} \bar{p} u_{t_1}^*) = \bar{p}$; that is, $\bar{p} \leq u_{t_1} \bar{p} u_{t_1}^*$.

Consequently, for the standing pattern, in $\mathcal{B}$: $\beta(\iota(\lambda))P = P$ for all $\lambda \in \Lambda$, and $P \leq \beta(t)P\beta(t)^*$.

Proof. (1) is the image under $(\iota_1)_*$ of the identity $u_\lambda p_{\Lambda_1} = p_{\Lambda_1}$ of Fact 2.2, since $(\iota_1)_*(u_\lambda) = u_{\iota_1(\lambda)}$.

For (2), both sides may be evaluated in an arbitrary unitary representation π of H_1 on a Hilbert space $\mathcal{H}$, since $C_{\max}^*(H_1)$ admits a faithful one. The composition $\pi \circ (\iota_1)_*$ is the integrated form of the unitary representation $\pi \circ \iota_1$ of Λ_1 (both are $*$ -homomorphisms extending the same unitary representation), so by Fact 2.2, $\pi(\bar{p})$ is the orthogonal projection onto

$$\text{Fix}(\pi \circ \iota_1) = \{\xi \in \mathcal{H} : \pi(\iota_1(\lambda))\xi = \xi \text{ for all } \lambda \in \Lambda_1\},$$

and, since $\pi(u_{t_1} \bar{p} u_{t_1}^*) = \pi(t_1) \pi(\bar{p}) \pi(t_1)^*$, the left-hand side is the orthogonal projection onto $\pi(t_1) \text{Fix}(\pi \circ \iota_1)$. We claim $\text{Fix}(\pi \circ \iota_1) \subseteq \pi(t_1) \text{Fix}(\pi \circ \iota_1)$. Let $\xi \in \text{Fix}(\pi \circ \iota_1)$ and put $\eta = \pi(t_1)^{-1} \xi$. For $\lambda \in \Lambda_1$ choose $\lambda' \in \Lambda_1$ with $t_1 \iota_1(\lambda) t_1^{-1} = \iota_1(\lambda')$; then

$$\pi(\iota_1(\lambda)) \eta = \pi(t_1)^{-1} \pi(t_1 \iota_1(\lambda) t_1^{-1}) \xi = \pi(t_1)^{-1} \pi(\iota_1(\lambda')) \xi = \pi(t_1)^{-1} \xi = \eta,$$

so $\eta \in \text{Fix}(\pi \circ \iota_1)$ and $\xi = \pi(t_1) \eta$. The inclusion of ranges gives $\pi(\bar{p}) \leq \pi(u_{t_1} \bar{p} u_{t_1}^*)$, i.e. (2) in the representation π ; as π was arbitrary and some π is faithful, (2) holds in $C_{\max}^*(H_1)$. The final statement follows by applying the $*$ -homomorphism $\tilde{\beta}$ to the instance $(H_1, \Lambda_1, \iota_1, t_1) = (H, \Lambda, \iota, t)$, for which (M1) is the required one-sided containment. $\square$

Lemma 7.11 (compression equality). $P = \beta(t) P \beta(t)^*$ in $\mathcal{B}$.

Proof. Put $Q_0 = \beta(t) P \beta(t)^*$ and $v = \beta(t) P$. Then $v^* v = P \beta(t)^* \beta(t) P = P$ and $vv^* = Q_0$, and $P \leq Q_0$ by Lemma 7.10. Apply Lemma 6.3(2) with $p := P$, $q := Q_0$, and the element v^* , which satisfies $(v^*)^*(v^*) = vv^* = Q_0$ and $(v^*)(v^*)^* = v^*v = P$; it yields $P = Q_0$. $\square$

We now pass from the corona projection P to its literal fixed-space action on a Hilbert-space ultraproduct. Fix a free ultrafilter ω on the infinite index set J' , and let

$$K_\omega = \left\{ (\xi_n)_{n \in J'} : \xi_n \in K_n, \sup_n \|\xi_n\| < \infty \right\} / \left\{ (\xi_n) : \lim_{n \rightarrow \omega} \|\xi_n\| = 0 \right\},$$

the Hilbert-space ultraproduct with inner product $\langle [\xi], [\eta] \rangle = \lim_{n \rightarrow \omega} \langle \xi_n, \eta_n \rangle$.

Lemma 7.12. *For a bounded sequence $(T_n)_{n \in J'}$ of operators with class $[T] \in \mathcal{B}$, the assignment $\sigma([T])[(\xi_n)] = [(T_n \xi_n)]$ is a well-defined unital $*$ -homomorphism $\sigma : \mathcal{B} \rightarrow B(K_\omega)$. Moreover $\sigma(P)$ is the orthogonal projection onto*

$$\text{Fix}_\omega = \{ \zeta \in K_\omega : \sigma(\beta(\iota(\lambda))) \zeta = \zeta \text{ for all } \lambda \in \Lambda \}.$$

Proof. If (T_n) is a null sequence in operator norm, then

$$\|T_n \xi_n\| \leq \|T_n\| \sup_n \|\xi_n\| \rightarrow 0.$$

In particular this holds along ω , so σ is well defined on $\mathcal{B}$, and it is clearly a unital $*$ -homomorphism. The map

$$\sigma \circ \tilde{\beta} \circ \iota_* : C_{\max}^*(\Lambda) \rightarrow B(K_\omega)$$

is a $*$ -representation on a Hilbert space extending the unitary representation $\lambda \mapsto \sigma(\beta(\iota(\lambda)))$; by uniqueness of integrated forms and Fact 2.2,

$$\sigma(P) = \sigma(\tilde{\beta}(\iota_*(p_\Lambda)))$$

is the orthogonal projection onto the fixed space of that representation, which is Fix_ω . $\square$

8. PROOF OF THE CRITERION AND OF THEOREM A

Proof of Theorem 7.2. Suppose, for contradiction, that some corona representation $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ is nontrivial on F , and adopt all notation fixed in Section 7: the corner representation Θ_- with $\sum_{f \in F} \Theta_-(f) = 0$, the unitary lifts $U_{g,n}$, the corona $\mathcal{B}$, the homomorphism β , the projection P , the ultraproduct K_ω and the representation σ .

Step 1: the lamp vector is fixed. Each $U_{c,n}$ is a unitary matrix, hence a vector in K_n of norm $\|U_{c,n}\|_2 = 1$; let $\xi_c = [(U_{c,n})] \in K_\omega$, a unit vector. For $\lambda \in \Lambda$ we have $\iota(\lambda) c \iota(\lambda)^{-1} = c$ in H by (M2), so $U_{\iota(\lambda),n} U_{c,n} U_{\iota(\lambda),n}^*$ and $U_{c,n}$ both lift $\Theta_-(c)$, whence $\|U_{\iota(\lambda),n} U_{c,n} U_{\iota(\lambda),n}^* - U_{c,n}\| \rightarrow 0$ and a fortiori the K_n -norms tend to 0. Therefore $\sigma(\beta(\iota(\lambda)))\xi_c = \xi_c$ for every $\lambda \in \Lambda$, i.e. $\xi_c \in \text{Fix}_\omega$, and by Lemma 7.12, $\sigma(P)\xi_c = \xi_c$.

Step 2: the translated lamp vector is fixed. Let $\xi_d := \sigma(\beta(t))\xi_c = [(D_n)]$, where $D_n := U_{t,n} U_{c,n} U_{t,n}^*$, an operator-norm lift of $\Theta_-(d)$. By Lemma 7.11 and Step 1,

$$\sigma(P)\xi_d = \sigma(\beta(t)P\beta(t)^*)\sigma(\beta(t))\xi_c = \sigma(\beta(t))\sigma(P)\xi_c = \xi_d,$$

so $\xi_d \in \text{Fix}_\omega$: for every $\lambda \in \Lambda$,

$$(9) \quad \lim_{n \rightarrow \omega} \|U_{\iota(\lambda),n} D_n U_{\iota(\lambda),n}^* - D_n\|_2 = 0.$$

Step 3: every defect commutator collapses in Hilbert–Schmidt norm. For $\lambda \in \Lambda$ put

$$X_{\lambda,n} := D_n U_{\iota(\lambda),n} D_n^{-1} U_{\iota(\lambda),n}^{-1},$$

a unitary in $M_{r_n}(\mathbb{C})$ and an operator-norm lift of $\Theta_-([d, \iota(\lambda)])$. Two applications of unitary invariance of $\|\cdot\|_2$ give

$$\|X_{\lambda,n} - 1\|_2 = \|D_n U_{\iota(\lambda),n} - U_{\iota(\lambda),n} D_n\|_2 = \|D_n - U_{\iota(\lambda),n} D_n U_{\iota(\lambda),n}^{-1}\|_2,$$

so by (9),

$$(10) \quad \lim_{n \rightarrow \omega} \|X_{\lambda,n} - 1\|_2 = 0 \quad \text{for every } \lambda \in \Lambda.$$

Step 4: the collapse propagates to the compression-defect subgroup. Let $f \in F$. Since $F \subseteq N_{\text{comp}}$, there are finitely many (possibly zero) elements $h_1, \dots, h_L \in H$, $\lambda_1, \dots, \lambda_L \in \Lambda$ and signs $\varepsilon_1, \dots, \varepsilon_L \in \{\pm 1\}$ with

$$f = \prod_{i=1}^L h_i [d, \iota(\lambda_i)]^{\varepsilon_i} h_i^{-1}.$$

Then

$$V_{f,n} := \prod_{i=1}^L U_{h_i,n} X_{\lambda_i,n}^{\varepsilon_i} U_{h_i,n}^*$$

is an operator-norm lift of $\Theta_-(f)$, and the telescoping estimate $\|AB - 1\|_2 \leq \|A - 1\|_2 + \|B - 1\|_2$ for unitaries A, B , together with unitary invariance ($\|UXU^* - 1\|_2 = \|X - 1\|_2$ and $\|X^{-1} - 1\|_2 = \|X - 1\|_2$), gives

$$\|V_{f,n} - 1\|_2 \leq \sum_{i=1}^L \|X_{\lambda_i,n} - 1\|_2 \xrightarrow{n \rightarrow \omega} 0$$

by (10).

Step 5: the corner average forbids the collapse. Each $V_{f,n}$ lifts $\Theta_-(f)$, so the finite sum $\sum_{f \in F} V_{f,n}$ is an operator-norm lift of $\sum_{f \in F} \Theta_-(f) = 0$:

$$\left\| \sum_{f \in F} V_{f,n} \right\| \longrightarrow 0 \quad (n \rightarrow \infty \text{ in } J'),$$

hence also along ω . On the other hand, by Step 4,

$$\left\| \sum_{f \in F} V_{f,n} - |F| \right\|_2 \leq \sum_{f \in F} \|V_{f,n} - 1\|_2 \xrightarrow{n \rightarrow \omega} 0,$$

so, using $\|x\|_2 \leq \|x\|$,

$$|F| = \||F| \cdot 1\|_2 \leq \left\| \sum_{f \in F} V_{f,n} - |F| \right\|_2 + \left\| \sum_{f \in F} V_{f,n} \right\| \xrightarrow{n \rightarrow \omega} 0,$$

which is absurd since $|F| \geq 1$. This contradiction proves Theorem 7.2. $\square$

Lemma 8.1 (the marked word is a defect square). *Let H_1 be a group and $d, a_1 \in H_1$ with $d^2 = 1$. Then*

$$[d, a_1 d a_1^{-1}] = [d, a_1]^2.$$

Proof. Put $b = a_1 d a_1^{-1}$, so $b^2 = a_1 d^2 a_1^{-1} = 1$; thus $b^{-1} = b$ and $d^{-1} = d$. Then $[d, a_1] = d a_1 d^{-1} a_1^{-1} = d(a_1 d a_1^{-1}) = db$, and

$$[d, a_1]^2 = (db)^2 = db db = d b d^{-1} b^{-1} = [d, b]. \quad \square$$

Proof of Theorem A. Part (1) is Proposition 4.3. For part (2), let $\iota: \Gamma \rightarrow E$ be the canonical homomorphism and $a = v_1 = (e_1, 1)$. The tuple (Γ, ι, t, c) is a marked Kazhdan pattern in E : Γ has property (T) by Fact 2.1; (M1) holds because $t \iota(\gamma) t^{-1} = \iota(\alpha(\gamma))$ for every $\gamma \in \Gamma$ (both sides are homomorphisms in γ , and (6) gives the identity on the six base generators); (M2) holds because $[c, g] = 1$ in (6) for the six base generators g . In E we have $d^2 = (t c t^{-1})^2 = t c^2 t^{-1} = 1$, so Lemma 8.1 with $a_1 = \iota(a)$ gives

$$w = [d, \iota(a) d \iota(a)^{-1}] = [d, \iota(a)]^2 \in N_{\text{comp}},$$

since $[d, \iota(a)]$ is one of the defining generators of N_{comp} . The subgroup $F = \{1, w\} = \langle w \rangle$ is finite and normal: centrality and $w^2 = 1$ are defining relations, while Proposition 4.3 gives $w \neq 1$. Moreover $F \subseteq N_{\text{comp}}$. Theorem 7.2

therefore gives $\Theta(w) = 1$ for every corona representation Θ of E , which is part (2).

For the final assertions of Theorem A: an MF embedding of E would be an injective corona representation, which is impossible because every corona representation sends the nontrivial element w to 1. If $C_{\text{red}}^*(E)$ were an MF algebra, a unital C^* -embedding — Corollary 9.5 below reduces a non-unital one to this case — $C_{\text{red}}^*(E) \hookrightarrow \mathcal{Q}$ composed with the canonical map $E \rightarrow \mathcal{U}(C_{\text{red}}^*(E))$, would exhibit an injective corona representation of E : the map $g \mapsto \lambda_g$ is injective, although the quotient map $C_{\text{max}}^*(E) \rightarrow C_{\text{red}}^*(E)$ need not be faithful. The canonical map $E \rightarrow \mathcal{U}(C_{\text{max}}^*(E))$ is also injective, since its composition with the regular representation is $g \mapsto \lambda_g$; this treats $C_{\text{max}}^*(E)$. $\square$

Remark 8.2. The proof shows more than non-injectivity of each individual Θ : the fixed element w lies in the kernel of *every* corona representation of E , so every corona representation factors through $E/\langle w \rangle$ (note $\langle w \rangle = \{1, w\}$ is central, hence normal). In the language of approximation radicals, w is a nontrivial element of the “MF radical” of E . This terminology is meant in analogy with Slofstra’s earlier quotients G^{fin} and G^{fa} , obtained by killing elements invisible to exact and approximate finite-dimensional representations, respectively [Slo17, Definitions 2.5–2.7]. By Theorem 7.2 the same holds in every countable group with a marked Kazhdan pattern: every finite normal subgroup of the compression-defect subgroup lies in the MF radical. Section 9 develops the radical systematically; we do not know whether $E/\langle w \rangle$ is MF, i.e. whether the MF radical of E is exactly $\{1, w\}$.

Remark 8.3. No lower bound on the corner ranks r_n , or on their density in d_n , is used anywhere: Lemma 6.5 renormalizes the trace so that all subsequent estimates are relative to the corner. This is what makes the operator-norm setting insensitive to small relative ranks, which block the same argument in Hilbert–Schmidt metrics (Section 11).

9. CONSEQUENCES

Throughout this section, “corona representation” and “MF” are as in Section 2.

9.1. The MF radical.

Definition 9.1. For a countable group G_1 , the *MF radical* is

$$\text{Rad}_{\text{MF}}(G_1) = \bigcap \{\ker \Theta : \Theta \text{ a corona representation of } G_1\},$$

a normal subgroup of G_1 . This is the operator-norm analogue of the invisible subgroups of [Slo17, Definitions 2.5–2.7] and of the approximation kernels studied in [NST].

Lemma 9.2 (portability). *For every homomorphism $\varphi: G_1 \rightarrow G_2$ of countable groups,*

$$\varphi(\text{Rad}_{\text{MF}}(G_1)) \subseteq \text{Rad}_{\text{MF}}(G_2):$$

the MF radical is functorial. In particular, if $\varphi: E \rightarrow G_2$ satisfies $\varphi(w) \neq 1$, then G_2 is not MF.

Proof. If $x \in \text{Rad}_{\text{MF}}(G_1)$ and Θ is a corona representation of G_2 , then $\Theta \circ \varphi$ is a corona representation of G_1 , so $\Theta(\varphi(x)) = 1$. For the second statement: $w \in \text{Rad}_{\text{MF}}(E)$ by Theorem A, so $1 \neq \varphi(w) \in \text{Rad}_{\text{MF}}(G_2)$, and an injective corona representation of G_2 would kill $\varphi(w)$, a contradiction. $\square$

Proposition 9.3 (the largest MF quotient). *Let G_1 be a countable group and $R = \text{Rad}_{\text{MF}}(G_1)$.*

(1) *R is the kernel of a single corona representation of G_1 .*

(2) *G_1/R is MF.*

(3) *Every homomorphism from G_1 to an MF group factors through G_1/R .*

In particular G_1 is MF if and only if $R = \{1\}$. The statement is presumably folklore once the radical is defined; we include a proof for completeness and claim no novelty for it.

Proof. (3) If $\varphi: G_1 \rightarrow M$ with M MF and $j: M \rightarrow \mathcal{U}(\mathcal{Q})$ is an injective corona representation, then $j \circ \varphi$ is a corona representation of G_1 , so $j(\varphi(R)) = 1$ and, by injectivity of j , $\varphi(R) = 1$.

(1) and (2). If $R = G_1$, the trivial representation has kernel R and G_1/R is trivial, hence MF; so assume $R \neq G_1$ and let $(g_i)_{i \geq 1}$ list, possibly with repetitions, elements of $G_1 \setminus R$ meeting every nontrivial coset of R . For each i choose a corona representation $\Theta_i: G_1 \rightarrow \mathcal{U}(\mathcal{Q}_i)$, $\mathcal{Q}_i = \mathcal{Q}((d_n^{(i)})_{n \in J_i})$, with $\Theta_i(g_i) \neq 1$; put $\varepsilon_i = \frac{1}{2} \|\Theta_i(g_i) - 1\| > 0$, and choose unitary lifts $(U_{g,n}^{(i)})_{n \in J_i}$ of $\Theta_i(g)$ for every $g \in G_1$ (Lemma 6.1), with $U_{1,n}^{(i)} = 1$. Recall that the quotient norm in a corona is the limit superior of the coordinate norms: subtracting a null sequence does not change the limit superior, and truncating finitely many coordinates realizes it.

Enumerate $G_1 = \{h_1, h_2, \dots\}$ and fix a stage $k \geq 1$. For each $i \leq k$: the finitely many products $U_{g,n}^{(i)} U_{h,n}^{(i)}$ and $U_{gh,n}^{(i)}$ with $g, h \in \{h_1, \dots, h_k\}$ lift the same corona elements, so there is n_0 with all the corresponding defects at most $1/k$ for $n \geq n_0$; and since $\limsup_{n \in J_i} \|U_{g_i,n}^{(i)} - 1\| = 2\varepsilon_i$, there are infinitely many $n \in J_i$ with $\|U_{g_i,n}^{(i)} - 1\| \geq \varepsilon_i$. Choose $n_i(k) \in J_i$ satisfying both conditions and set

$$\Phi_k(g) = \bigoplus_{i \leq k} U_{g, n_i(k)}^{(i)} \in \mathcal{U}(M_{D_k}(\mathbb{C})), \quad D_k = \sum_{i \leq k} d_{n_i(k)}^{(i)}.$$

Define $\Theta_\infty(g)$ as the class of $(\Phi_k(g))_{k \in \mathbb{N}}$ in $\mathcal{Q}((D_k)_{k \in \mathbb{N}})$. For fixed $g, h \in G_1$, both lie in $\{h_1, \dots, h_k\}$ for all large k , and the operator norm of a block sum is the maximum of the block norms, so $\|\Phi_k(g)\Phi_k(h) - \Phi_k(gh)\| \leq 1/k \rightarrow 0$;

together with $\Phi_k(1) = 1$ this makes Θ_∞ a corona representation of G_1 . Consequently $\Theta_\infty(R) = 1$. Conversely, if $g \notin R$, pick i with $g \in g_i R$; then $\Theta_\infty(g) = \Theta_\infty(g_i)$, and for every $k \geq i$, $\|\Phi_k(g_i) - 1\| \geq \|U_{g_i, n_i(k)}^{(i)} - 1\| \geq \varepsilon_i$, so $\|\Theta_\infty(g_i) - 1\| \geq \varepsilon_i$ and $\Theta_\infty(g) \neq 1$. Hence $\ker \Theta_\infty = R$, proving (1), and Θ_∞ descends to an injective corona representation of G_1/R , proving (2). For the final equivalence: if G_1 is MF, an injective corona representation forces $R = \{1\}$; if $R = \{1\}$, then (2) says $G_1 = G_1/R$ is MF. $\square$

Corollary 9.4 (exact radical from a candidate quotient). *Let $N \trianglelefteq G_1$ satisfy $N \leq \text{Rad}_{\text{MF}}(G_1)$. Every corona representation of G_1 factors uniquely through G_1/N . If in addition G_1/N is MF, then*

$$\text{Rad}_{\text{MF}}(G_1) = N.$$

Thus proving that a concrete defect subgroup is invisible and that its quotient is MF computes both the entire MF radical and the largest MF quotient.

Proof. The containment $N \leq \ker \Theta$ gives the unique quotient factorization for every corona representation Θ . For the reverse containment in the displayed equality, compose the quotient map $G_1 \rightarrow G_1/N$ with an injective corona representation of G_1/N : its kernel is exactly N , while the MF radical lies in the kernel of every corona representation. $\square$

9.2. Targets beyond coronas.

Corollary 9.5 (subalgebras of coronas). *Let H , (Λ, ι, t, c) and $F \subseteq N_{\text{comp}}$ be as in Theorem 7.2. Let A be a nonzero C^* -subalgebra of a norm matrix corona $\mathcal{Q}$ whose unit $p := 1_A$ is not necessarily the unit of $\mathcal{Q}$, and let $\Theta: H \rightarrow \mathcal{U}(A)$ be a group homomorphism. Then $\Theta(f) = 1_A$ for every $f \in F$. The same conclusion holds when A is a unital separable MF algebra — for instance a unital separable quasidiagonal algebra (Fact 2.4), in particular an AF algebra, or $C_{\text{red}}^*(F_k)$ for a free group F_k [HT, Corollary 8.4].*

Proof. Every unitary u of A satisfies $u = pup$ and $u^*u = uu^* = p$, so $\mathcal{U}(A) \subseteq \mathcal{U}(p\mathcal{Q}p)$, and $p \neq 0$. By Lemma 6.4 there is a unital $*$ -isomorphism of $p\mathcal{Q}p$ onto a norm matrix corona carrying p to the unit. Composing, Θ becomes a corona representation of H , and Theorem 7.2 gives $\Theta(f) = 1_A$ for $f \in F$. If A is a unital separable MF algebra, fix a $*$ -monomorphism $\psi: A \rightarrow \mathcal{Q}_1$ into a norm matrix corona, unital or not [BK]; then $\psi(A)$ is a C^* -subalgebra of $\mathcal{Q}_1$ with unit $\psi(1_A)$, and applying the first part to $\psi \circ \Theta$ gives $\psi(\Theta(f)) = \psi(1_A)$, whence $\Theta(f) = 1_A$ by injectivity. Unital separable quasidiagonal algebras embed unitaly into coronas by Fact 2.4, AF algebras are quasidiagonal, and $C_{\text{red}}^*(F_k)$ is MF for every $k \in \mathbb{N} \cup \{\infty\}$ by [HT, Corollary 8.4]. $\square$

Corollary 9.6. *The group E admits no injective homomorphism into the unitary group of any C^* -subalgebra of any norm matrix corona — in particular, none into $\mathcal{U}(A)$ for A a unital separable MF algebra, a unital separable quasidiagonal algebra, or $C_{\text{red}}^*(F_k)$.*

Proof. By the proof of Theorem A, the pattern (Γ, ι, t, c) and $F = \{1, w\}$ satisfy the hypotheses of Corollary 9.5 in $H = E$; hence any such homomorphism kills w , while $w \neq 1$ by Proposition 4.3. $\square$

9.3. Stable finiteness does not imply MF.

Lemma 9.7. *Let G_1 be a countable discrete group.*

- (1) *The canonical tracial state τ on $C_{\text{red}}^*(G_1)$ is faithful.*
- (2) *Every unital C^* -algebra A with a faithful tracial state is stably finite: for every $k \geq 1$, every isometry in $M_k(A)$ is a unitary.*

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Proof. (1) Realize $C_{\text{red}}^*(G_1) \subseteq B(\ell^2 G_1)$ as the norm closure of the span of the left translation unitaries λ_g , and let ρ_g be the right translation unitaries, $(\rho_g \xi)(h) = \xi(hg)$; each ρ_g commutes with each $\lambda_{g'}$, hence with every element of $C_{\text{red}}^*(G_1)$. The canonical trace is $\tau(x) = \langle x\delta_e, \delta_e \rangle$. If $\tau(x^*x) = 0$ then $\|x\delta_e\| = 0$, so $x\delta_e = 0$, and then $x\delta_g = x\rho_{g^{-1}}\delta_e = \rho_{g^{-1}}x\delta_e = 0$ for every $g \in G_1$. The vectors δ_g are total in $\ell^2 G_1$, so $x = 0$.

(2) On $M_k(A)$ the functional $\tau_k([x_{ij}]) = \frac{1}{k} \sum_i \tau(x_{ii})$ is a state with the trace property $\tau_k(xy) = \frac{1}{k} \sum_{i,j} \tau(x_{ij}y_{ji}) = \tau_k(yx)$, and it is faithful: the i -th diagonal entry of x^*x is $\sum_j x_{ji}^*x_{ji}$, so $\tau_k(x^*x) = 0$ forces $\tau(x_{ji}^*x_{ji}) = 0$, hence $x_{ji} = 0$, for all i, j . If $v \in M_k(A)$ satisfies $v^*v = 1$, then $y := 1 - vv^*$ is a projection with $\tau_k(y) = \tau_k(1) - \tau_k(vv^*) = \tau_k(1) - \tau_k(v^*v) = 0$; since $y = y^*y$, faithfulness gives $y = 0$, i.e. $vv^* = 1$. $\square$

Proof of Theorem D. $C_{\text{red}}^*(E)$ is unital and separable since E is countable; by Lemma 9.7 it carries a faithful tracial state and is stably finite. It is not an MF algebra by Theorem A. $\square$

Remark 9.8. Blackadar and Kirchberg proved that MF algebras are stably finite [BK]. The general converse was already refuted as a consequence of the negative solution to the Connes embedding problem [Go23, Corollary 5.2]; Theorem D instead realizes that failure as the reduced group C^* -algebra of an explicit finitely presented group, with a faithful trace. The example is not nuclear: Γ embeds into E (Corollary 4.5); the infinite group Γ has property (T) and is therefore not amenable, since amenable groups with property (T) are compact, hence finite when discrete [BHV, Theorem 1.1.6]; and the reduced C^* -algebra of a nonamenable group is not nuclear [BO, Theorem 2.6.8]. The nuclear territory is untouched: whether every stably finite separable nuclear C^* -algebra is quasidiagonal remains open, and by [TWW] every faithful trace on a separable nuclear C^* -algebra satisfying the UCT is quasidiagonal.

9.4. The ambient HNN extension.

Lemma 9.9 (residually finite groups are MF). *Every countable residually finite group is MF.*

Proof. Let G_1 be countable and residually finite; if G_1 is trivial the statement is trivially true, so enumerate its nontrivial elements $x_1, x_2, \dots$ (with

repetitions if G_1 is finite). For each k choose a finite-index normal subgroup $N_k \trianglelefteq G_1$ with $x_k \notin N_k$, put $M_k = N_1 \cap \cdots \cap N_k$, a finite-index normal subgroup, and let $\varphi_k: G_1 \rightarrow G_1/M_k$ be the quotient map onto the finite group G_1/M_k . Let π_k be the left regular representation of G_1/M_k composed with φ_k , a finite-dimensional unitary representation of G_1 . Define $\Theta(g)$ as the class of $(\pi_k(g))_k$ in $\mathcal{Q}((|G_1/M_k|)_k)$; each π_k is an exact representation, so Θ is a corona representation. For $g = x_i$ and $k \geq i$ the element $\varphi_k(x_i)$ is nontrivial, so the permutation matrix $\pi_k(x_i)$ moves some standard basis vector δ , giving $\|(\pi_k(x_i) - 1)\delta\| = \sqrt{2}$ and $\|\pi_k(x_i) - 1\| \geq \sqrt{2}$. Hence $\|\Theta(x_i) - 1\| \geq \sqrt{2}$ and Θ is injective. $\square$

Proposition 9.10 (the ambient group). *Let G be the ascending HNN extension (5).*

- (1) $C_{\max}^*(G)$ contains an explicitly described nonunitary isometry. In particular $C_{\max}^*(G)$ is not finite, a fortiori not stably finite, and not an MF algebra.
- (2) G is residually finite, hence MF.

Proof. (1) Write $\iota_G: \Gamma \rightarrow G$ for the canonical homomorphism and $\bar{p} = (\iota_G)_*(p_\Gamma) \in C_{\max}^*(G)$ for the image of the Kazhdan projection. The data (G, Γ, ι_G, t) satisfy the hypotheses of Lemma 7.10, because $t \iota_G(\gamma) t^{-1} = \iota_G(\alpha(\gamma))$; hence $\bar{p} \leq Q_1 := u_t \bar{p} u_t^*$. Put

$$s = \bar{p} u_t^* + (1 - Q_1).$$

From $\bar{p} \leq Q_1$ we get $\bar{p} Q_1 = Q_1 \bar{p} = \bar{p}$, hence $(1 - Q_1) \bar{p} = 0$ and $\bar{p}(1 - Q_1) = 0$; also $Q_1 u_t \bar{p} = u_t \bar{p} u_t^* u_t \bar{p} = u_t \bar{p}$, so $(1 - Q_1) u_t \bar{p} = 0$ and, taking adjoints, $\bar{p} u_t^* (1 - Q_1) = 0$. Therefore

$$\begin{aligned} s^* s &= u_t \bar{p} \bar{p} u_t^* + u_t \bar{p} (1 - Q_1) + (1 - Q_1) \bar{p} u_t^* + (1 - Q_1) \\ &= Q_1 + (1 - Q_1) = 1 \end{aligned}$$

and

$$\begin{aligned} s s^* &= \bar{p} u_t^* u_t \bar{p} + \bar{p} u_t^* (1 - Q_1) + (1 - Q_1) u_t \bar{p} + (1 - Q_1) \\ &= \bar{p} + (1 - Q_1) = 1 - (Q_1 - \bar{p}) : \end{aligned}$$

s is an isometry with defect $1 - s s^* = Q_1 - \bar{p}$. Thus $s^* s = 1$ makes s left-invertible, with left inverse s^* , whereas $s s^* \neq 1$ makes s a nonunit. Equivalently, with $a = s^*$ and $b = s$ one has $ab = 1$ but $ba \neq 1$, so $C_{\max}^*(G)$ is not directly finite.

It remains to check $Q_1 \neq \bar{p}$. Let $H_0 = t \iota_G(\Gamma) t^{-1} \leq G$ and let π be the quasi-regular representation of G on $\ell^2(G/H_0)$, with basepoint $b = H_0 \in G/H_0$. As in the proof of Lemma 7.10, $\pi(\bar{p})$ is the orthogonal projection onto the $\iota_G(\Gamma)$ -invariant vectors, and $\pi(Q_1) = \pi(t) \pi(\bar{p}) \pi(t)^*$. The vector $\pi(t)^{-1} \delta_b = \delta_{t^{-1} H_0}$ is $\iota_G(\Gamma)$ -invariant: for $\gamma \in \Gamma$,

$$\iota_G(\gamma) t^{-1} H_0 = t^{-1} (t \iota_G(\gamma) t^{-1}) H_0 = t^{-1} H_0,$$

since $t \iota_G(\gamma) t^{-1} \in H_0$; hence $\pi(Q_1) \delta_b = \delta_b$. On the other hand $\pi(\iota_G(a)) \delta_b = \delta_{a H_0} \neq \delta_b$: if $a H_0 = H_0$ then $a \in H_0$, and applying the homomorphism

$G \rightarrow \bar{G}$ of Lemma 4.2, which restricts to the identity on Γ and sends t to D , would give $a \in D\Gamma D^{-1} = \alpha(\Gamma)$ inside $\mathrm{GL}_4(\mathbb{Z}[1/2])$, contradicting Lemma 3.1. So δ_b is not $\iota_G(\Gamma)$ -invariant, i.e. $\pi(\bar{p})\delta_b \neq \delta_b$ while $\pi(Q_1)\delta_b = \delta_b$; hence $Q_1 \neq \bar{p}$ and s is not a unitary.

If $C_{\max}^*(G)$ embedded unittally into a norm matrix corona (Corollary 9.5 reduces a non-unital MF embedding to this case), the image of s would be an isometry there, hence a unitary by Lemma 6.3(1), and injectivity would force $ss^* = 1$, a contradiction; so $C_{\max}^*(G)$ is not an MF algebra, and the isometry s itself witnesses the failure of finiteness.

(2) G is the mapping torus of the injective endomorphism α of the finitely generated linear group $\Gamma \leq \mathrm{GL}_4(\mathbb{Z})$, so G is residually finite by Borisov–Sapir [BoS, Theorem 1.6]; MF-ness follows from Lemma 9.9. $\square$

Remark 9.11. Proposition 9.10 separates group-level and C^* -level MF-ness in the direction opposite to Theorem D: G is an MF group whose maximal group C^* -algebra is not even finite, while E is a non-MF group whose reduced group C^* -algebra is stably finite. The one-way implication — if $C_{\max}^*$ or C_{red}^* of a group is an MF algebra then the group is MF — is recorded in [Sh].

9.5. Failure of permanence properties.

Corollary 9.12 (quotients). *MF groups are not closed under quotients: the free group F_8 is MF, and E is a quotient of F_8 .*

Proof. E is generated by eight elements (6), hence is a quotient of F_8 . Free groups are residually finite: F_8 embeds into F_2 (a subgroup of index 7 of F_2 is free of rank 8 by the classical Nielsen–Schreier theorem), F_2 embeds into $\mathrm{SL}_2(\mathbb{Z})$ by Sanov’s classical ping-pong pair $(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}), (\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix})$, residual finiteness passes to subgroups, and $\mathrm{GL}_2(\mathbb{Z})$ is residually finite by reduction modulo primes (a nontrivial integer matrix stays nontrivial modulo any prime exceeding the absolute values of the entries of its difference from the identity). So F_8 is MF by Lemma 9.9, while E is not MF by Theorem A. $\square$

Proposition 9.13 (the witness group is non-MF). *Let $W = \mathrm{Clamp}(\bar{X}) \rtimes \bar{G}$ be the witness group of Proposition 4.3. Then W is finitely generated, the homomorphism $E \rightarrow W$ of that proof is surjective, and the central sign $z = -1 \in \mathrm{Clamp}(\bar{X})$ satisfies $z \in \mathrm{Rad}_{\mathrm{MF}}(W) \setminus \{1\}$. In particular W is not MF.*

Proof. The image of the homomorphism $E \rightarrow W$ contains the six base generators of Γ , hence Γ ; it contains D ; so it contains $\bar{G} = \langle \Gamma, D \rangle$. It contains c_o , hence $c_g\Gamma = g c_o g^{-1}$ for every $g \in \bar{G}$, which is all of $\{c_x : x \in \bar{X}\}$ since $\bar{G}$ acts transitively on $\bar{X} = \bar{G}/\Gamma$. Finally $(c_{D\Gamma} c_{aD\Gamma})^2 = -c_{D\Gamma}^2 c_{aD\Gamma}^2 = -1 = z$ because the two sites are distinct (Lemma 4.2), so the image contains z and therefore all of $\mathrm{Clamp}(\bar{X}) = \langle \pm 1, c_x : x \in \bar{X} \rangle$: the map is surjective, and W is generated by the images of the eight generators of E . The proof of Proposition 4.3 sends w to z , so Lemma 9.2 gives $z \in \mathrm{Rad}_{\mathrm{MF}}(W)$, and $z \neq 1$ in W by Construction 4.1. A group whose MF radical is nontrivial admits no injective corona representation. $\square$

Corollary 9.14 (extensions). *MF groups are not closed under extensions, even split ones: in the split exact sequence*

$$1 \longrightarrow \text{Cilamp}(\bar{X}) \longrightarrow W \longrightarrow \bar{G} \longrightarrow 1,$$

the kernel is locally finite, hence amenable and MF; the quotient $\bar{G} \leq \text{GL}_4(\mathbb{Z}[1/2])$ is a finitely generated linear group, hence residually finite by Mal'cev's theorem [Ma] and MF by Lemma 9.9; and W is not MF (Proposition 9.13). In particular there is a finitely generated non-MF group that is (locally finite)-by-(residually finite). This contrasts with the positive permanence theory for MF groups in [Sh].

Proof. Exactness and splitting hold by construction of the semidirect product. Every finite subset of $\text{Cilamp}(\bar{X})$ lies in the subgroup generated by ± 1 and finitely many c_x , a finite 2-group, as in the proof of Theorem C; so $\text{Cilamp}(\bar{X})$ is locally finite, hence amenable, hence MF by Facts 2.3 and 2.4. The remaining assertions were proved above. $\square$

Corollary 9.15 (a simple envelope). *There exists a 2-generator simple group S with $\text{Rad}_{\text{MF}}(S) = S$: every corona representation of S , and every homomorphism from S into any MF group, is trivial.*

Proof. By Goryushkin's theorem [Go], the countable group E embeds into a 2-generator simple group S . By Lemma 9.2 applied to the embedding, the image of w is a nontrivial element of $\text{Rad}_{\text{MF}}(S)$. The radical is a normal subgroup of the simple group S , so $\text{Rad}_{\text{MF}}(S) = S$, i.e. every corona representation of S is trivial. If $\varphi: S \rightarrow M$ is a homomorphism to an MF group and j an injective corona representation of M , then $j \circ \varphi$ is a corona representation, hence trivial, so φ is trivial. The simple-envelope theorem is entirely Goryushkin's; what is added here is the nontrivial portable radical element. $\square$

9.6. Finitely presented Kazhdan envelopes. We use the following recent external embedding theorem. Chatterji and Kassabov prove that, given a finitely generated recursively presented group H and $1 \neq h \in H$, there is an embedding $j: H \hookrightarrow \Lambda$ into a finitely presented property (T) group Λ which is normally generated by $j(h)$ [CK, Theorem 3]. The arXiv record [arXiv:2601.22907v1](https://arxiv.org/abs/2601.22907v1) was submitted on January 30, 2026. We use only that theorem and make no claim that its construction is part of the present work.

Theorem 9.16 (a finitely presented Kazhdan group with full MF radical). *There exist a finitely presented property (T) group Λ and an embedding $j: E \hookrightarrow \Lambda$ such that*

$$\text{Rad}_{\text{MF}}(\Lambda) = \Lambda.$$

Every corona representation of Λ , and every homomorphism from Λ to an MF group, is trivial. Moreover every finite-dimensional linear representation of Λ , over every field, is trivial, and Λ is perfect.

Proof. Apply [CK, Theorem 3] to (E, w) . Portability gives $j(w) \in \text{Rad}_{\text{MF}}(\Lambda)$. The radical is normal, whereas $j(w)$ normally generates Λ , so the radical is all of Λ . The statements about corona representations and maps to MF groups follow from Proposition 9.3.

Let $\pi: \Lambda \rightarrow \text{GL}(V)$ with V finite-dimensional over an arbitrary field. Theorem B, applied to $\pi \circ j$, gives $\pi(j(w)) = 1$; normal generation gives $\pi = 1$. Finally w is a commutator, hence $j(w)$ lies in the commutator subgroup of Λ . That normal subgroup contains a normal generator, so it is all of Λ and Λ is perfect. $\square$

In particular Λ has no nontrivial finite or residually finite quotient. It also has no nontrivial amenable quotient and no nontrivial finitely generated linear quotient: quotients of the finitely generated group Λ are countable, amenable groups are MF by Facts 2.3 and 2.4, and finitely generated linear groups are residually finite by Mal'cev's theorem [Ma]. Thus Λ is minimally almost periodic. By the classical Peter–Weyl theorem [PW], its Bohr compactification is trivial.

Theorem 9.17 (a universal Kazhdan group with full MF radical). *There exists a finitely presented property (T) group Λ_{univ} which contains every finitely presented group as a subgroup and satisfies*

$$\text{Rad}_{\text{MF}}(\Lambda_{\text{univ}}) = \Lambda_{\text{univ}}.$$

Every homomorphism from Λ_{univ} to an MF group is trivial.

Proof. Enumerate all finite group presentations and form the free product of the groups they present. The Higman–Neumann–Neumann and Higman embedding theorems [HNN, Hig] embed this recursively presented countable group into a finitely presented group U . Thus U contains every finitely presented group. Choose $i: E \hookrightarrow U$. Portability and injectivity give $1 \neq i(w) \in \text{Rad}_{\text{MF}}(U)$. Apply [CK, Theorem 3] to $(U, i(w))$. Its image in the resulting finitely presented property (T) envelope normally generates, so portability forces the final MF radical to be the entire group. Universality survives the embedding of U , and the last assertion follows from Proposition 9.3. $\square$

The directionality is worth emphasizing: Λ_{univ} contains every finite group and every finitely presented residually finite, amenable, or linear group, while admitting no nontrivial quotient in any of those MF classes.

Corollary 9.18 (opposite full and reduced behavior in one group). *For the same universal Kazhdan group,*

$$C_{\text{red}}^*(\Lambda_{\text{univ}}) \text{ is stably finite but non-MF,}$$

whereas $C_{\text{max}}^(\Lambda_{\text{univ}})$ contains a proper isometry and is therefore neither finite nor stably finite, and hence is not MF.*

Proof. The canonical trace on $C_{\text{red}}^*(\Lambda_{\text{univ}})$ is faithful, so the proof of Lemma 9.7, applied to every finite matrix amplification, gives stable finiteness. If the reduced algebra embedded faithfully in a norm matrix corona — unittally, by Corollary 9.5 — restriction to its left-regular unitaries would give an injective corona representation of Λ_{univ} , contradicting its full MF radical.

The finitely presented ambient HNN group G of Proposition 9.10 embeds in Λ_{univ} . For $H \leq K$, the canonical map $C_{\text{max}}^*(H) \rightarrow C_{\text{max}}^*(K)$ is injective: induce any unitary representation of H to K and recover the original representation on the identity-coset summand, so the maximal norm is preserved. Thus the explicit proper isometry in $C_{\text{max}}^*(G)$ remains proper in $C_{\text{max}}^*(\Lambda_{\text{univ}})$. A proper isometry witnesses failure of finiteness and stable finiteness. $\square$

9.7. The scaling family. For an integer $m \geq 2$ let E_m be the group with the presentation (6), except that the three relations $tv_it^{-1} = v_i^2$ are replaced by

$$tv_it^{-1} = v_i^m \quad (1 \leq i \leq 3);$$

thus $E = E_2$, while the marked word $w_m = [tct^{-1}, v_1(tct^{-1})v_1^{-1}]$ is unchanged.

Theorem 9.19 (scaling family). *For every $m \geq 2$: the group E_m is finitely presented, w_m is a nontrivial central involution in E_m , and every corona representation of E_m kills w_m . Consequently E_m is not MF, neither $C_{\text{max}}^*(E_m)$ nor $C_{\text{red}}^*(E_m)$ is an MF algebra, and $C_{\text{red}}^*(E_m)$ is stably finite but not MF.*

Proof. Finite presentation is clear from the displayed presentation.

The pattern. Define $\alpha_m: \Gamma \rightarrow \Gamma$ by $\alpha_m(v, A) = (mv, A)$. The computation of Lemma 3.1 applies with 2 replaced by m : α_m is an injective endomorphism with image $m\mathbb{Z}^3 \rtimes \text{SL}_3(\mathbb{Z})$, and $a = (e_1, 1) \notin \alpha_m(\Gamma)$ because $e_1 \notin m\mathbb{Z}^3$. On the six base generators α_m fixes x, y, z and sends v_i to v_i^m , so the relations of E_m assert $t\gamma t^{-1} = \alpha_m(\gamma)$ on generators; hence, with $\iota_m: \Gamma \rightarrow E_m$ the canonical homomorphism, $t\iota_m(\gamma)t^{-1} = \iota_m(\alpha_m(\gamma))$ for all $\gamma \in \Gamma$, and c commutes with $\iota_m(\Gamma)$: the tuple (Γ, ι_m, t, c) is a marked Kazhdan pattern in E_m (Fact 2.1). As in the proof of Theorem A, $d^2 = tc^2t^{-1} = 1$ in E_m , and Lemma 8.1 gives $w_m = [d, \iota_m(a)]^2 \in N_{\text{comp}}$; the subgroup $\{1, w_m\}$ is finite and normal, so Theorem 7.2 kills w_m in every corona representation.

Nontriviality. Let $D_m = \text{diag}(m, m, m, 1) \in \text{GL}_4(\mathbb{Z}[1/m])$. As in Lemma 4.2, $D_m \begin{pmatrix} A & v \\ 0 & 1 \end{pmatrix} D_m^{-1} = \begin{pmatrix} A & mv \\ 0 & 1 \end{pmatrix}$ is the matrix of $\alpha_m(v, A)$, and $D_m^{-1}aD_m = \begin{pmatrix} 1 & \frac{1}{m}e_1 \\ 0 & 1 \end{pmatrix} \notin \Gamma$, so in $\bar{G}_m := \langle \Gamma, D_m \rangle \leq \text{GL}_4(\mathbb{Z}[1/m])$ the cosets $D_m\Gamma$ and $aD_m\Gamma$ of $\bar{X}_m := \bar{G}_m/\Gamma$ are distinct. The proof of Proposition 4.3 applies with $(\bar{G}, \bar{X}, D)$ replaced by $(\bar{G}_m, \bar{X}_m, D_m)$ — its only inputs were the conjugation identity, the fixing of the base coset by Γ , and the coset separation — and sends w_m to $[c_{D_m\Gamma}, c_{aD_m\Gamma}] = -1 \neq 1$ in $\text{CILamp}(\bar{X}_m) \rtimes \bar{G}_m$.

The final assertions follow exactly as in the proofs of Theorems A and D, with E_m in place of E . $\square$

Remark 9.20. We do not claim the groups E_m are pairwise non-isomorphic; multiplicity is addressed in Corollary 9.22. For every $d \geq 3$ the same construction runs with $\mathbb{Z}^d \rtimes \mathrm{SL}_d(\mathbb{Z})$ in place of Γ — property (T) holds by the same references [BHV, Sections 1.4–1.5 and 4.2] — once a finite presentation of the base is fixed; we confine the formal statement to $d = 3$, where the presentation (4) is completely explicit.

9.8. Multiplicity.

Lemma 9.21. *Every subgroup of an MF group is MF; equivalently, a group containing a non-MF subgroup is not MF.*

Proof. The restriction of an injective corona representation to a subgroup is an injective corona representation. $\square$

Corollary 9.22 (multiplicity).

- (1) *The groups $E \times \mathbb{Z}^k$, $k \in \mathbb{N}$, are pairwise non-isomorphic, finitely presented, and non-MF.*
- (2) *There exist $2^{\aleph_0}$ pairwise non-isomorphic finitely generated non-MF groups.*

Proof. Every group in sight contains an isomorphic copy of E , hence is non-MF by Theorem A and Lemma 9.21.

(1) $E \times \mathbb{Z}^k$ is finitely presented because E is. Its abelianization is $E^{\mathrm{ab}} \times \mathbb{Z}^k$; since E is finitely generated, E^{ab} is a finitely generated abelian group, of some torsion-free rank $r_0 < \infty$, so the abelianization of $E \times \mathbb{Z}^k$ has torsion-free rank $r_0 + k$, an isomorphism invariant that separates distinct k .

(2) By a classical construction of B. H. Neumann [Neu] there is a family $(N_i)_{i \in I}$ with $|I| = 2^{\aleph_0}$ of pairwise non-isomorphic 2-generator groups. Put $G_i = E \times N_i$, a finitely generated non-MF group. For a fixed countable group G_1 , the set $\{i \in I : G_i \cong G_1\}$ is countable: a countable group has only countably many finitely generated subgroups (each is determined by a finite generating subset), hence countably many isomorphism classes of such; if $G_i \cong G_1$ then N_i embeds into G_1 as $\{1\} \times N_i$; and the N_i are pairwise non-isomorphic, so $i \mapsto [N_i]$ is injective on the set in question. A map from a set of cardinality $2^{\aleph_0}$ with countable fibres has image of cardinality $2^{\aleph_0}$; apply this to $i \mapsto [G_i]$. $\square$

9.9. Algorithmic unrecognizability. We use the Adian–Rabin theorem in its standard Markov-property form [Ad58, Rab]: if an isomorphism-invariant property has a finitely presented positive example and a finitely presented forbidden group which cannot embed in any finitely presented group having the property, then the property is undecidable from finite presentations.

Theorem 9.23 (MF recognition is undecidable). *There is no algorithm which, given a finite group presentation, decides whether the presented group is MF.*

Proof. Being MF is inherited by subgroups (Lemma 9.21). The trivial group is a finitely presented positive example, while the finitely presented non-MF group E cannot embed in any MF group. Thus MF is a Markov property, and Adian–Rabin applies. $\square$

Theorem 9.24 (negative instances are not recursively enumerable). *For any standard effective coding of finite group presentations, the set of codes presenting non-MF groups is not recursively enumerable. Consequently there is no sound and complete recursively verifiable system of finite certificates for non-MF-ness of finitely presented groups.*

Proof. Fix a finitely presented group N with undecidable word problem [No, Boo]. The effective test-group construction in the proof of Adian–Rabin (see [LS, Section IV.4]) associates to every word u a finite presentation P_u such that $u = 1$ in N implies $G(P_u) = 1$, whereas $u \neq 1$ implies that E embeds in $G(P_u)$. Hence $G(P_u)$ is MF exactly when $u = 1$. Trivial words are recursively enumerable. If non-MF finite presentations were also recursively enumerable, pulling that enumeration back along $u \mapsto P_u$ would enumerate the nontrivial words of N ; running both enumerations in parallel would decide its word problem, a contradiction. The certificate statement follows because the objects accepted by a recursive finite verifier form a recursively enumerable set. $\square$

Corollary 9.25 (C^* -recognition problems). *There is no algorithm which, from a finite presentation of a group H , decides any of the following four properties:*

- (1) $C_{\text{red}}^*(H)$ is MF;
- (2) $C_{\text{max}}^*(H)$ is MF;
- (3) $C_{\text{max}}^*(H)$ is finite (equivalently, for a unital C^* -algebra, directly finite);
- (4) $C_{\text{max}}^*(H)$ is stably finite.

For each property, the set of finite presentations on which it fails is not recursively enumerable.

Proof. For $H \leq K$, the restriction of the left regular representation of K to H is a direct sum of copies of the left regular representation of H , so $C_{\text{red}}^*(H) \hookrightarrow C_{\text{red}}^*(K)$. The analogous maximal embedding was recalled in the proof of Corollary 9.18. MF-ness, finiteness, and stable finiteness pass to unital C^* -subalgebras. Thus each property is subgroup-hereditary at the group level.

For the first two properties, the forbidden finitely presented group is E . For the last two it is the finitely presented group G from Proposition 9.10, whose maximal algebra contains a proper isometry. The trivial group is a positive example in every case, so the Adian–Rabin theorem gives undecidability. Applying its effective test-group construction to the respective forbidden group gives the same pullback argument as in Theorem 9.24, hence failure of recursive enumerability on the negative side. $\square$

9.10. Uniform invisibility.

Theorem 9.26 (uniform invisibility). *Let H , (Λ, ι, t, c) and $F \subseteq N_{\text{comp}}$ be as in Theorem 7.2, and fix $f_0 \in F$ — for instance $H = E$ and $f_0 = w$. For every $\varepsilon > 0$ there exist $\delta > 0$ and a finite subset $F_0 \subseteq H$ such that for every $r \geq 1$ and every map $\varphi: H \rightarrow \mathcal{U}(M_r(\mathbb{C}))$ with*

$$\|\varphi(g)\varphi(h) - \varphi(gh)\| \leq \delta \quad \text{for all } g, h \in F_0,$$

one has $\|\varphi(f_0) - 1\| < \varepsilon$.

Proof. Suppose the statement fails for some $\varepsilon_0 > 0$. Fix an enumeration $H = \{h_1, h_2, \dots\}$ (with repetitions if H is finite) and put $F_k = \{h_1, \dots, h_k\}$. For each $k \geq 1$, the failure at $(\delta, F_0) = (1/k, F_k)$ produces $r_k \geq 1$ and $\varphi_k: H \rightarrow \mathcal{U}(M_{r_k}(\mathbb{C}))$ with $\|\varphi_k(g)\varphi_k(h) - \varphi_k(gh)\| \leq 1/k$ for all $g, h \in F_k$, but $\|\varphi_k(f_0) - 1\| \geq \varepsilon_0$. Define $\Theta(g)$ as the class of the sequence $(\varphi_k(g))_{k \in \mathbb{N}}$ in $\mathcal{Q}((r_k)_{k \in \mathbb{N}})$, a unitary element. For fixed $g, h \in H$ we have $g, h \in F_k$ for all large k , so the displayed defect tends to 0 and $\Theta(g)\Theta(h) = \Theta(gh)$. Taking $g = h = 1$ gives $\|\varphi_k(1)^2 - \varphi_k(1)\| \leq 1/k$ for large k , hence $\|\varphi_k(1) - 1\| \leq 1/k$ after multiplying by the unitary $\varphi_k(1)^*$; so $\Theta(1) = 1$, and then $\Theta(g)\Theta(g^{-1}) = \Theta(1) = 1$ for every g . Thus Θ is a corona representation of H , and Theorem 7.2 gives $\Theta(f_0) = 1$, i.e. $\|\varphi_k(f_0) - 1\| \rightarrow 0$ — contradicting $\|\varphi_k(f_0) - 1\| \geq \varepsilon_0$. $\square$

Remark 9.27. Conversely, the uniform statement applied to unitary lifts recovers the kill part of Theorem 7.2, so the two are equivalent. The compactness proof yields no explicit (δ, F_0) , and none is claimed; an effective version would require tracking the Kazhdan constant of Λ through Sections 7–8, which we do not pursue.

The closedness assertion below is a standard consequence of Dadarlat’s finite-local characterization of MF groups and his observation that groups locally embeddable in MF groups are MF [Dad, Proposition 3.2 and Remark 3.3(i)], together with the standard fact that a marked limit locally embeds in its approximating groups; for a modern formulation see [FFGS, Theorem 2.16]. We claim no novelty for that implication. Korchagin [Kor, Propositions 5 and 7] records the equivalent uniform local criterion, with separation constant $\sqrt{2}$. We record it because the word-ball certificate is useful here; the full topological statement will also be used below.

Theorem 9.28 (marked-space closedness and the literal cylinder). *For every k , the MF locus in the space $\mathcal{G}_k$ of k -marked groups is closed; equivalently, the non-MF locus is open. More precisely, every non-MF marked group has a finite reduced-word ball whose relation pattern defines a neighborhood consisting entirely of non-MF groups.*

For the canonical eight-generator marking of E , there is the explicit clopen neighborhood

$$\mathcal{O}_E = \{(H; h_1, \dots, h_8) : r(h_1, \dots, h_8) = 1 \ (r \in R), \quad w(h_1, \dots, h_8) \neq 1\}.$$

Every group in $\mathcal{O}_E$ is non-MF. With the literal relator set of Definition 3.2, membership is determined by the relation pattern on reduced words of length at most 34.

Proof. Lemma 6.2, applied pairwise and assembled by block sums exactly as in the proof of Theorem 7.3, gives the following normalized local criterion: a countable group is MF exactly when, for every finite subset and every positive error, it has an operator-norm unitary model whose distinct elements are separated by at least 1.

Suppose $(G_n, S_n) \rightarrow (G, S)$ and every G_n is MF. Given a finite $F \subseteq G$, choose words in the free marked generators representing every element of $F \cup F^2$. On a sufficiently small cylinder, all equalities and inequalities among these finitely many words agree with those in G . A normalized local model in one such G_n therefore pulls back, word by word, to a local model of G with the same multiplicative error and separation. The criterion makes G MF. This proves closedness; recording the finitely many control words inside one reduced word ball gives the stronger finite-radius statement.

For the literal neighborhood, the relators give a surjection $E \twoheadrightarrow H$. If the marked word survives in H , portability of $w \in \text{Rad}_{\text{MF}}(E)$ puts its nontrivial image in $\text{Rad}_{\text{MF}}(H)$, so H is non-MF. Equality or inequality of each fixed word is clopen, and there are finitely many relators. Direct reduction of the displayed words gives length at most 16 for w and at most 34 for every relator (the longest are the centrality relators $[w, g]$), proving the asserted radius. $\square$

Remark 9.29 (open questions). Two questions remain adjacent to this section. (i) Is $E/\langle w \rangle$ MF, i.e. is the MF radical of E exactly $\{1, w\}$? (ii) Does a *torsion-free* non-MF group exist? Theorem 7.3 removes the former conceptual obstruction: it would suffice to construct a torsion-free finitely presented H with a nontrivial normal property (T) subgroup $K \leq N_{\text{comp}}$ (or, after the corresponding whole-defect upgrade, $K \leq \mathfrak{D}(H, L)$). A plausible route is to combine a torsion-free Rips construction, which supplies the normal Kazhdan kernel, with the one-sided-compressor geometry in Fournier-Facio’s torsion-free nonsofic construction [FF]; we do not carry out that construction here.

10. THE CYCLIC-BASE COMPARISON

We prove Theorem C. The broader phenomenon that a marked central involution can be invisible to every exact finite-dimensional representation but visible to approximate representations is central to Slofstra’s work [Slo17]; the point here is its occurrence in the same operator-norm marked-compression pattern as Theorem A. Recall the setting: $G_0 = \text{BS}(1, 2)$ with base $\Gamma_0 = \langle \gamma_0 \rangle \cong \mathbb{Z}$ and stable letter t satisfying $t\gamma_0 t^{-1} = \gamma_0^2$; $X_0 = G_0/\Gamma_0$; $E_0 = \text{CLamp}(X_0) \rtimes G_0$ with central sign $z_0 = -1 \in \text{CLamp}(X_0)$ as in Construction 4.1. Figure 4 summarizes the comparison.

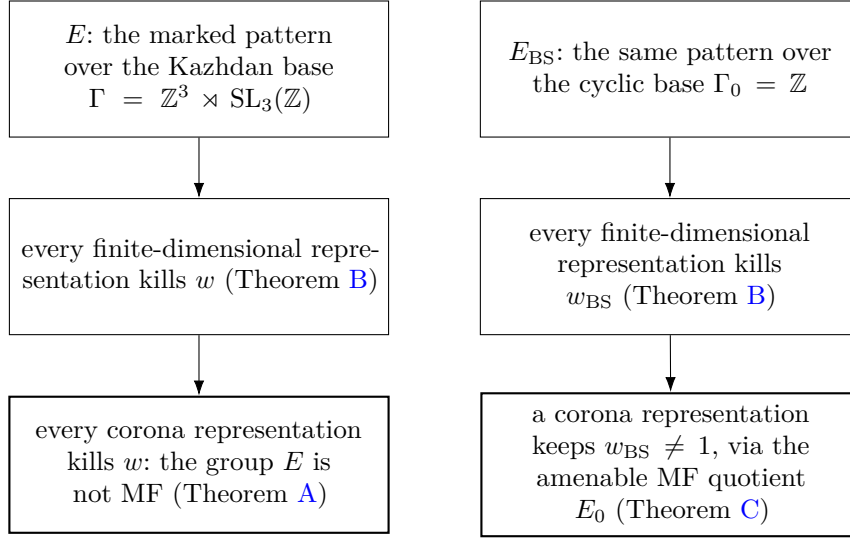


FIGURE 4. The cyclic-base comparison. The finite-dimensional obstruction does not see the base, while the corona conclusion holds exactly on the side where property (T) supplies the uniform fixed-space projection.

Proof of Theorem C. $z_0 \neq 1$: immediate from Construction 4.1, since $\text{ClLamp}(X_0) \leq \text{Cl}(X_0)^\times$ and $\text{Cl}(X_0) \neq 0$.

Amenability: every finitely generated subgroup of $\text{ClLamp}(X_0)$ is contained in the subgroup generated by ± 1 and finitely many c_x , which is a finite 2-group (it has order at most 2^{k+1} for k sites, since every element is $\pm a$ a reduced product of distinct generators). Hence $\text{ClLamp}(X_0)$ is locally finite, so amenable. The group $G_0 = \text{BS}(1, 2) \cong \mathbb{Z}[1/2] \rtimes \mathbb{Z}$ is metabelian, so amenable. As an extension of the amenable G_0 by the amenable $\text{ClLamp}(X_0)$, the group E_0 is amenable.

MF: by Facts 2.3 and 2.4, the amenable group E_0 is MF, and an MF embedding $E_0 \hookrightarrow \mathcal{U}(\mathcal{Q})$ is injective, so the image of z_0 is a nontrivial (self-adjoint) unitary in the corona: the central sign survives.

The quotient from the literal presentation: assign $\gamma_0 \mapsto \gamma_0$, $t \mapsto t$, and $c \mapsto c_o$, where $o = \Gamma_0 \in X_0$. The point o is fixed by Γ_0 , so $[c_o, \gamma_0] = 1$. Also $t\Gamma_0 t^{-1} = \langle \gamma_0^2 \rangle \subseteq \Gamma_0$. Moreover $\gamma_0 t \Gamma_0 \neq t \Gamma_0$ in G_0 : otherwise $t^{-1} \gamma_0 t \in \Gamma_0$; in the faithful affine model $\text{BS}(1, 2) = \mathbb{Z}[1/2] \rtimes \mathbb{Z}$ with $\gamma_0 = (1, 0)$ and $t = (0, 1)$ acting by multiplication by 2, we compute $t^{-1} \gamma_0 t = (\frac{1}{2}, 0) \notin \Gamma_0 = \mathbb{Z} \times \{0\}$. Consequently

$$w_{\text{BS}} \mapsto [c_{t\Gamma_0}, c_{\gamma_0 t \Gamma_0}] = z_0.$$

The image is central of order two, so every relator of E_{BS} is respected. The resulting homomorphism is surjective: γ_0, t generate G_0 , the conjugates of c_o give all c_x because G_0 acts transitively on X_0 , and their commutators

contain z_0 . Composing this quotient with the MF embedding of E_0 gives the asserted corona representation of E_{BS} .

The finite-dimensional obstruction persists: E_0 contains the pattern of Theorem B with $H = E_0$, $\Gamma = \Gamma_0$, $t = t$, $c = c_o$ ($o = \Gamma_0 \in X_0$, fixed by Γ_0), and $a = \gamma_0 \in \Gamma_0$. We have $t\Gamma_0 t^{-1} = \langle \gamma_0^2 \rangle \subseteq \Gamma_0$ and $[c_o, \Gamma_0] = 1$. Thus Theorem B says every finite-dimensional representation of E_0 , over every field, kills z_0 . Applying the same theorem directly to the displayed presentation of E_{BS} shows that every finite-dimensional representation kills w_{BS} . $\square$

Remark 10.1. Theorem C shows that the argument cannot be generalized merely by deleting its fixed-space rigidity input. The elementary Theorem B is genuinely weaker than the corona conclusion: E_0 has MF approximations even though its finite-dimensional representations all kill the sign.

11. LIMITATIONS: HILBERT–SCHMIDT METRICS AND HYPERLINEARITY

The proof of Theorem A does not extend to tracial ultraproducts, for a structural reason worth recording.

Suppose one replaces the norm corona $\mathcal{Q}$ by a tracial ultraproduct $\mathcal{M} = \prod^\omega (M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$, with defects measured in $\|\cdot\|_2$ rather than operator norm, and asks whether a hyperlinear-style embedding of E must kill w . Two steps break.

First, Lemma 7.9 fails: Hilbert–Schmidt-multiplicative lifts need not be operator-norm multiplicative, so the adjoint maps $\text{Ad } U_{g,n}$ need not assemble into a homomorphism into any norm corona. One can instead pass to the conjugation representation on $L^2(\mathcal{M})$, which is an honest representation of E .

The tracial ultraproduct $\mathcal{M}$ itself is a finite von Neumann algebra. Second, and decisively, Lemma 6.3 is unavailable after the argument passes to its conjugation action. The available ambient algebra is $B(L^2(\mathcal{M}))$; when $L^2(\mathcal{M})$ is infinite-dimensional this algebra is properly infinite and has no normalized finite trace with which to compare the fixed-space projections. Thus the inclusion supplied by the analogue of Lemma 7.10 is not forced to be equality. Proper inclusion between nested unitarily equivalent projections is possible in such an ambient algebra (the unilateral-shift phenomenon), but we do not claim that it occurs for a tracial representation of the present group. In particular the present paper proves nothing about hyperlinearity or Connes embeddability of E , and nothing about the sofic-versus-hyperlinear problem.

The dividing line in the present argument is thus: *operator-norm matrix models form a stably finite world, in which property (T) compression is rigid; for tracial matrix models the available ambient algebra of the conjugation action can be properly infinite, so the same projection comparison need not collapse to equality.*

This failure is closely related to a current structural question rather than to a known impossibility theorem. Alekseev–Thom ask whether, for a Kazhdan representation into a tracial ultraproduct of matrices, its commutant can be recovered as an ultraproduct of finite-dimensional subalgebras, possibly as coordinate centralizers of suitable lifts [AT, Open Problem 6.2]. Such a coordinate model would provide finite-level structure absent from the argument above, but their question is open and does not restore the present stable-finiteness proof in the tracial setting.

TOOL AND COMPUTATIONAL RESOURCE DISCLOSURE

The mathematics in this paper was developed and formalized with Anthropic’s Claude Fable and OpenAI’s GPT-5.6 Sol.

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